Journal of Cleaner Production 552 (2026) 147987

Contents lists available at ScienceDirect

# Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

# An integrated resilience dynamic assessment framework for urban road transportation systems under daily disruptions

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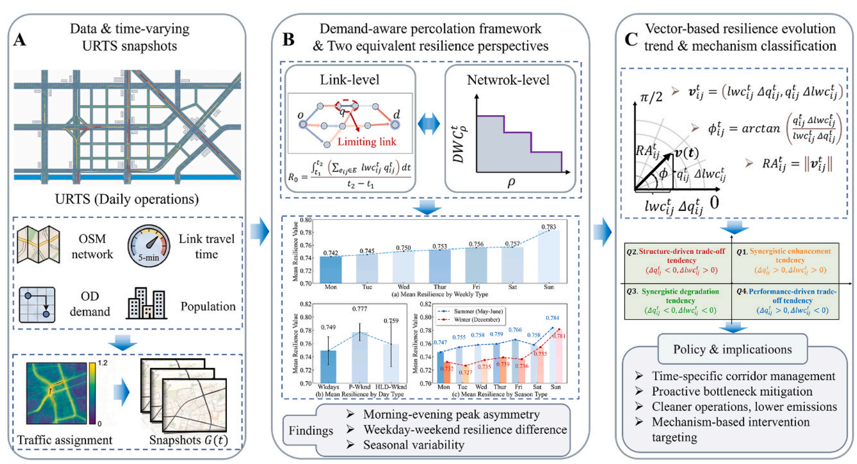
## HIGHLIGHTS

- Demand-aware, performance-based percolation framework for daily resilience analysis.
- Proving equivalence and mutual interpretability of network and link-level resilience.
- Proposing a vector-based resilience evolution trend classification method.
- Identifying a resilience recovery phenomenon under peak-hour pressure.

## GRAPHICAL ABSTRACT




## ARTICLE INFO

*Keywords:*
Daily resilience assessment
Demand-aware percolation
Vector-based trend characterization method
Resilience evolution trend map

## ABSTRACT

Urban road transportation systems (URTS) are subject to recurrent, low-impact disturbances that cause gradual functional degradation rather than abrupt failures. To assess resilience under such daily operating conditions, this study develops a demand-aware, performance-based percolation framework of observed daily traffic operations. It integrates link operating performance and demand-induced structural dependence within a unified scheme. Building on link performance and demand-weighting formulations, daily operational resilience is quantified through temporal aggregation over a time window of network states, enabling consistent assessment from both network-level and link-level perspectives. A theoretical equivalence between the demand-percolation formulation and the link-aggregated representation is established, connecting system-wide outcomes with local link contributions. To further reveal how resilience evolves over time and space, a vector-based trend characterization method is proposed by decomposing resilience variation into first-order components associated with

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https://doi.org/10.1016/j.jclepro.2026.147987
Received 18 October 2025; Received in revised form 1 March 2026; Accepted 10 March 2026
Available online 12 March 2026
0959-6526/

performance change and structural reconfiguration. The method quantifies both the magnitude and direction of link-level resilience evolution and supports a mechanism-oriented classification of evolutionary tendencies. A case study of Nanjing using high-resolution traffic data demonstrates systematic temporal variability across weekdays versus weekends, peak versus off-peak periods, seasonal conditions, and the weekday-weekend contrast is statistically significant. The evolution analysis further identifies a phenomenon that even when network-level resilience remains low under peak-hour pressure, some links already exhibit improving resilience trends. The proposed framework provides an interpretable basis for time-specific and corridor-focused traffic management, and it supports sustainability goals by targeting recurrent congestion that contributes to cumulative energy use and emissions.

## 1. Introduction

Urban road transportation systems (URTS) form the backbone of daily mobility in modern cities, and their efficient operation is a fundamental prerequisite for sustainable urban mobility (Pan et al., 2022; Wang et al., 2024; Zhang et al., 2024b). Improving traffic operation performance not only enhances the reliability and service quality of URTS, but also contributes to reducing resource consumption and environmental impacts associated with urban traffic congestion (Pamucar et al., 2021). As urban road networks in many cities have gradually matured, enhancing daily operational resilience has emerged as an important means of improving operational efficiency under recurrent traffic disturbances (Zhang et al., 2024a). Consequently, the development of dynamic, daily-oriented resilience assessment approaches has become a critical prerequisite for supporting operational optimization, informing traffic management and network improvement strategies toward more efficient and sustainable urban transportation operations (Bi et al., 2024; Knoester et al., 2024; Salvo et al., 2025).

Recently, existing studies have assessed transportation system resilience using topology-based approaches (Chen et al., 2023; Ilalokhoin et al., 2023; Xu et al., 2024b), attribute-based (Twumasi-Boakye and Sobanjo, 2018; Zhou et al., 2019), and performance-based approaches (Wang et al., 2023a; Yin et al., 2023; Knoester et al., 2024). However, most of these assessments are conducted under specific disruption scenarios or from single-level perspectives, with limited attention to resilience under daily operating conditions. Resilience assessment under disruption scenarios differs fundamentally from resilience assessment under daily operating conditions in terms of disturbance mechanisms. While extreme events are often characterized by abrupt failures and binary connectivity loss, daily operations involve gradual performance degradation and recovery, accompanied by continuous changes in travel time, capacity utilization, and demand-induced structural dependence (Ambühl et al., 2023; Zhu et al., 2025). These dynamics are difficult to capture using interruption-based or static resilience indicators. This limitation motivates the development of resilience assessment approaches tailored to continuous, recurrent daily traffic disturbances. In addition, resilience is often evaluated using single, time-aggregated indicators, limiting the ability to characterize its temporal evolution under continuously varying operating conditions.

In recent years, percolation theory has been increasingly applied in complex network research to characterize the progressive degradation of system functionality through the filtering of network components. This perspective is conceptually consistent with daily urban road operations, where road performance degrades gradually, and traffic demand is continuously redistributed rather than abruptly interrupted. Accordingly, percolation-based approaches provide a natural framework for representing continuous, congestion-driven changes in network functionality beyond binary "connected-failed" assumptions.

Motivated by these considerations, this study develops a demand-aware percolation-based framework to assess the daily operational resilience of URTS. By jointly considering road operational performance and structural dependence induced by travel demand within a unified analytical framework, the proposed approach aims to provide a more suitable foundation for examining resilience dynamics under recurrent daily traffic disturbances. Furthermore, to characterize the dynamic evolution of resilience across temporal and spatial dimensions, a vector-based method for characterizing resilience evolution trends is introduced to quantify both the direction and magnitude of resilience change at the link level. This method enables a mechanism-oriented interpretation of how performance variation and demand redistribution jointly shape link-level resilience across time and space. Collectively, this framework provides a new analytical perspective for understanding the dynamic mechanisms underlying the daily operational resilience of urban road transportation systems. The main contributions of this study are summarized as follows.

(1) A percolation-based, demand-aware framework is proposed to assess the daily operational resilience of urban road transportation systems. The framework integrates operational performance and structural dependence within a unified modeling scheme, enabling resilience analysis from both a network perspective and a link-level perspective. These two perspectives are theoretically equivalent and together capture link heterogeneity under daily traffic disturbances.
(2) A vector-based method for characterizing resilience evolution trends is proposed to capture the spatiotemporal dynamics of resilience under daily traffic disturbances. By decomposing resilience variation into performance-driven and structure-related components, the method provides a mechanism-oriented characterization of the direction and intensity of resilience evolution at the link level.

The organization of the paper is as follows: Section 2 reviews related studies and identifies research gaps. Section 3 presents the proposed methodology. The study area, data sources and assessment results are presented in Section 4, followed by a discussion and implications in Section5. Conclusions are given in Section 6.

## 2. Literature review

This section reviews existing studies from two perspectives: (1) methodological frameworks for transportation system resilience assessment, and (2) applications of percolation theory in resilience-related analyses.

### 2.1. Assessing the resilience of transportation networks

The concept of resilience was first introduced by Holling (1973) to describe a system's capacity to absorb disturbances and maintain essential functions. In transportation systems, resilience is commonly interpreted as the ability to resist, absorb, and recover from performance degradation under disruptions, often characterized using the resilience triangle framework (Wang et al., 2023b). This study follows this conceptual interpretation and applies it to the assessment of daily operational resilience.

Existing studies on URTS network resilience have developed a variety of quantitative assessment approaches, which can generally be classified into topology-based, attribute-based, and performance-based methods. Topology-based approaches rely on complex network theory and graph-based metrics to characterize system robustness and vulnerability from the perspective of connectivity and structural stability. For example, Osei-Asamoah and Lownes (2014) evaluated structural resilience using global efficiency and largest connected component metrics. Wei and Xu (2024) assessed road network stability by integrating multiple graph-based topological indicators. Boeing and Ha (2024) analyzed topological resilience emphasizing network connectivity and structurally critical bottlenecks. These methods are effective for identifying structurally critical nodes and links under facility failures, but they typically abstract traffic flows and operational dynamics, limiting their applicability to time-varying performance analysis.

Attribute-based approaches extend topological analysis by incorporating infrastructure characteristics such as road hierarchy, capacity, redundancy, criticality, and resource constraints, thereby enhancing the engineering interpretability of resilience assessments. Xu et al. (2018) quantified network redundancy using travel alternative diversity and residual capacity attributes. Barker et al. (2013) evaluated resilience using component-importance attributes reflecting link failure impacts. Chopra et al. (2016) assessed the robustness of rail transit systems under disruption scenarios, while Saadat et al. (2020) analyzed risk exposure of critical metro facilities using vulnerability-based frameworks. Although attribute-based methods provide valuable insights into how infrastructure characteristics influence resilience across different dimensions, their reliance on scenario-specific attribute selection and weighting makes it difficult to systematically capture the coupled temporal evolution of operational performance and demand-induced structural dependence.

Performance-based approaches directly evaluate resilience using operational outcomes such as travel time, delay, and capacity utilization. De-Los-Santos et al. (2012) assessed resilience using travel-time performance indicators under transit disruptions. Lu (2018) quantified resilience via a gravity-based performance metric incorporating flows and recovery speed. Tang et al. (2021) defined rail transit resilience based on time-series performance loss functions derived from commuter travel times, while Chen et al. (2022) incorporated passenger behavioral responses into performance-based resilience assessments. Performance-based methods are closely linked to travel reliability (Pan et al., 2022), operational efficiency, and sustainability outcomes (Ahern, 2011), and have therefore received considerable attention. However, they are predominantly applied to event-driven or disaster-related disruptions, limiting their ability to capture the continuous, heterogeneous performance fluctuations that characterize daily traffic operations (Ma et al., 2023).

By contrast, research on travel time reliability has demonstrated that high-frequency, low-impact demand fluctuations constitute a primary source of daily travel uncertainty in urban road networks. Mattsson and Jenelius (2015) showed that recurrent demand variability plays a dominant role in shaping daily network performance distributions, while Carrion and Levinson (2012) emphasized the importance of accounting for routine travel time fluctuations when evaluating network-level performance. Subsequent studies have further developed reliability-based modeling and evaluation methods to quantify performance variability under daily operating conditions. Wang et al. (2017) highlighted the role of reliability indicators in capturing stability variations in road network operations. Although daily disturbances are moderate in isolation, their cumulative effects can degrade network efficiency and exacerbate energy consumption and emissions, thereby exerting sustained pressure on urban transportation sustainability (Sánchez-Balseca et al., 2023). Accordingly, assessing resilience from a daily operational perspective is essential for understanding both traffic efficiency and long-term sustainability under routine congestion.

### 2.2. Applications of percolation theory in transportation resilience

Percolation theory was originally developed to characterize abrupt connectivity transitions and critical threshold behaviors in random systems as occupation probabilities vary (Zhu et al., 2024). With the development of complex network theory, percolation models have been increasingly applied to transportation networks to analyze system degradation processes (Yao and Chen, 2023). In transportation applications, percolation processes are typically implemented by defining removal rules for nodes or links and gradually increasing threshold values to simulate the degradation of network elements, thereby revealing critical deterioration states and structurally important bottlenecks.

Recent studies have demonstrated that percolation theory is effective in capturing key transitions in urban traffic systems. Ambühl et al. (2023) linked percolation with macroscopic fundamental diagrams to analyze congestion-induced connectivity transitions. Zeng et al. (2020) identified multistable traffic states using percolation and GPS data under connectivity disruptions. Kwon et al. (2025) quantified percolation thresholds of free-flow clustering and spatial fragmentation in urban road networks. These studies collectively show that percolation critical points are closely associated with traffic bottlenecks, connectivity loss, and functional degradation under varying traffic conditions. Beyond congestion-oriented analyses, percolation-based approaches have also been applied to transportation networks under extreme disturbance scenarios. For example, Fan et al. (2020) incorporated percolation processes into flood propagation models to analyze the evolution of road network connectivity under flooding events, illustrating the applicability of percolation theory for resilience assessment under severe disruptions. Collectively, these studies indicate that percolation frameworks can represent progressive functionality loss rather than instantaneous collapse, a property that is also relevant for daily traffic disturbances.

In recent years, several studies have explored the application of percolation-related concepts to daily operational analysis in urban transportation systems, particularly under high-frequency, low-impact traffic disturbances. Liu et al. (2025) developed a multi-criteria resilience assessment framework for urban rail transit stations under daily operational disturbances. Their study combined station-level performance indicators with a two-stage passenger flow redistribution model, enabling the identification of vulnerable stations under routine demand fluctuations. This work provides valuable insights into station-level operational resilience; however, its analytical scope is confined to localized facilities and does not address network-wide connectivity dynamics within a unified percolation framework. Moreover, Hamedmoghadam et al. (2021) proposed a percolation-based framework that explicitly incorporates heterogeneous flow demand to identify bottleneck links governing network flow dynamics under congestion. By progressively filtering links based on congestion-induced performance degradation, their approach demonstrated that mitigating a small subset of critical bottlenecks can yield substantial system-wide efficiency gains. While this framework establishes an important theoretical foundation for demand-aware percolation analysis, it is primarily oriented toward bottleneck identification and snapshot-based network efficiency assessment, rather than the continuous evolution of resilience under daily operating conditions.

Existing percolation-based studies of transportation networks have largely concentrated on bottleneck detection or structural connectivity analysis within individual snapshots. Under daily operating conditions, performance degradation and demand redistribution co-evolve and jointly reshape demand-induced structural dependence. However, these coupled dynamics are seldom integrated into a single percolation-based resilience framework. Consequently, a systematic approach that links system-level outcomes with link-level functional contributions while capturing spatiotemporal resilience evolution remains limited.

### 2.3. Research gaps

Table 1 summarizes representative studies on transportation system

**Table 1**
Comparison of representative studies on transportation system resilience.

| Reference | Research target | | | Disturbance type | | Analysis scale | | | Resilience evolution | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | URTS | UPTS | MNTS | DD | SD | N | L | S | Yes | Partial | No |
| Tang et al. (2020) | ✓ | | | ✓ | | ✓ | | | ✓ | | |
| Hamedmoghadam et al. (2021) | ✓ | | | ✓ | | | ✓ | | | | ✓ |
| Pan et al. (2022) | ✓ | | | ✓ | | ✓ | | | | | ✓ |
| Liu et al. (2022) | | | ✓ | | ✓ | ✓ | | | | ✓ | |
| Ilalokhoin et al. (2023) | | ✓ | | | ✓ | ✓ | | | | ✓ | |
| Yin et al. (2023) | ✓ | | | | ✓ | ✓ | | | | ✓ | |
| Wang et al. (2023b) | | | ✓ | | ✓ | ✓ | | | | | ✓ |
| Wei and Xu (2024) | ✓ | | | | ✓ | ✓ | | | | ✓ | |
| Zhang et al. (2024a) | | | ✓ | | ✓ | ✓ | | | | | ✓ |
| Bi et al. (2024) | | ✓ | | | ✓ | ✓ | | | | | ✓ |
| Xu et al. (2024a) | | ✓ | | | ✓ | ✓ | | ✓ | | | ✓ |
| Zhang et al. (2024b) | ✓ | | | | ✓ | | ✓ | | | ✓ | |
| Chen et al. (2024) | | ✓ | | | ✓ | ✓ | | | | | ✓ |
| Ma et al. (2024) | | | ✓ | | ✓ | ✓ | | | | | ✓ |
| Liu et al. (2025) | | ✓ | | ✓ | | | | ✓ | | | ✓ |
| Ashja-Ardalan et al. (2025) | ✓ | | | | ✓ | ✓ | | | | | ✓ |
| Zhu et al. (2025) | | ✓ | | ✓ | | ✓ | | | | ✓ | |
| Ma et al. (2025) | | | ✓ | | ✓ | ✓ | | | | | ✓ |
| This study | ✓ | | | ✓ | | ✓ | ✓ | | ✓ | | |

Note: The serious disruptions considered in this review refer to disturbances caused by natural disasters or extreme weather events. Abbreviations: URTS, urban road transportation system; UPTS, urban public transportation system; MNTS, multi-modal transportation system; DD, daily disturbance; SD, serious disruption; N, network-level; L, link-level; S, station-level. Resilience evolution analysis indicates whether and how temporal evolution is examined: *Yes* denotes explicit analysis of continuous or repeated evolution patterns; *Partial* denotes stage- or scenario-based analyses without a unified evolution framework; *No* denotes single-event or static assessments.

resilience across disturbance contexts, analytical scales, and consideration of resilience evolution. Overall, based on the above review, two major research gaps can be identified:

(1) Existing studies on transportation system resilience have predominantly focused on serious, low-frequency disruption scenarios, while relatively limited attention has been paid to daily operational resilience under recurrent, low-impact traffic disturbances. Moreover, daily resilience assessments are often conducted at a single analytical scale, typically at the network level, with insufficient consideration of link-level heterogeneity in operational performance, demand-induced structural contribution, and how these local dynamics collectively shape system-wide resilience outcomes.
(2) Most existing resilience assessment frameworks rely on scenario-based indicators, providing limited insight into how resilience evolves continuously over time and space. In particular, there is a lack of systematic framework to characterize the direction, magnitude, and typical patterns of resilience evolution under daily operating conditions, which constrains the understanding of underlying resilience mechanisms and limits their practical value for adaptive traffic management.

## 3. Methodology

To address the gaps in existing research, this study proposes a percolation-based, demand-aware assessment framework for evaluating the daily operational resilience of URTS. The framework is designed for retrospective assessment using historical traffic observations and is complemented by a vector-based method for characterizing link-level resilience evolution trends under daily traffic disturbances. Fig. 1 outlines the overall framework of this study. Multi-source urban data are first processed to generate 15-min OD demand and time-dependent network snapshots $G(t)$. Daily operational resilience is then evaluated from two equivalent perspectives: the link-level and network-level. Finally, a vector-based method characterizes link-level resilience evolution and classifies typical evolutionary tendencies, enabling mechanism-oriented diagnosis.

### 3.1. Percolation-based modeling for daily traffic disturbances

Percolation theory has been used to characterize the progressive degradation of system functionality through the filtering of network components (Hamedmoghadam et al., 2021; Ma et al., 2025). In this study, a link-based percolation model is adopted, where the percolation process operates on the operational state of road segments. Specifically, percolation is conducted on time-dependent network snapshots at time interval $t$. The operational condition of each link $(i,j)$ at time interval $t$ is quantified by a link performance $q_{ij}^t$, defined as

$$q_{ij}^t = \frac{\min_{t'} \tau_{ij}^{t'}}{\tau_{ij}^t} \tag{1}$$

where $\tau_{ij}^t$ and $\min_{t'} \tau_{ij}^{t'}$ denote the average travel time at time interval $t$ and the daily minimum travel time on link $(i,j)$, respectively. Lower values of $q_{ij}^t$ indicate more severe congestion, and this definition constitutes a performance indicator derived from observed traffic operations. A performance threshold $\rho \in [0,1]$ is introduced to determine whether a link remains functionally active in the percolated network. Specifically, the percolation rule is defined as

$$\phi_{e_{ij}}^t = \begin{cases} 1, & q_{ij}^t \geq \rho, \\ 0, & q_{ij}^t < \rho, \end{cases} \tag{2}$$

where $\phi_{e_{ij}}^t$ denotes the connection status of link $(i,j)$ at time interval $t$. This rule represents graded functional degradation under daily disturbances rather than identifying a single critical threshold.

### 3.2. Construction of URTS network

To capture temporal variations in traffic operations and demand distribution under daily disturbances, the urban road transportation system is represented as a sequence of time-dependent network snapshots. For each time interval $t$, the URTS network snapshot is represented as a graph $G(t) = (V, E, A, F^t, Q^t)$. Here $V$ and $E$ denote the sets of nodes and links, respectively. Nodes represent physical road in-

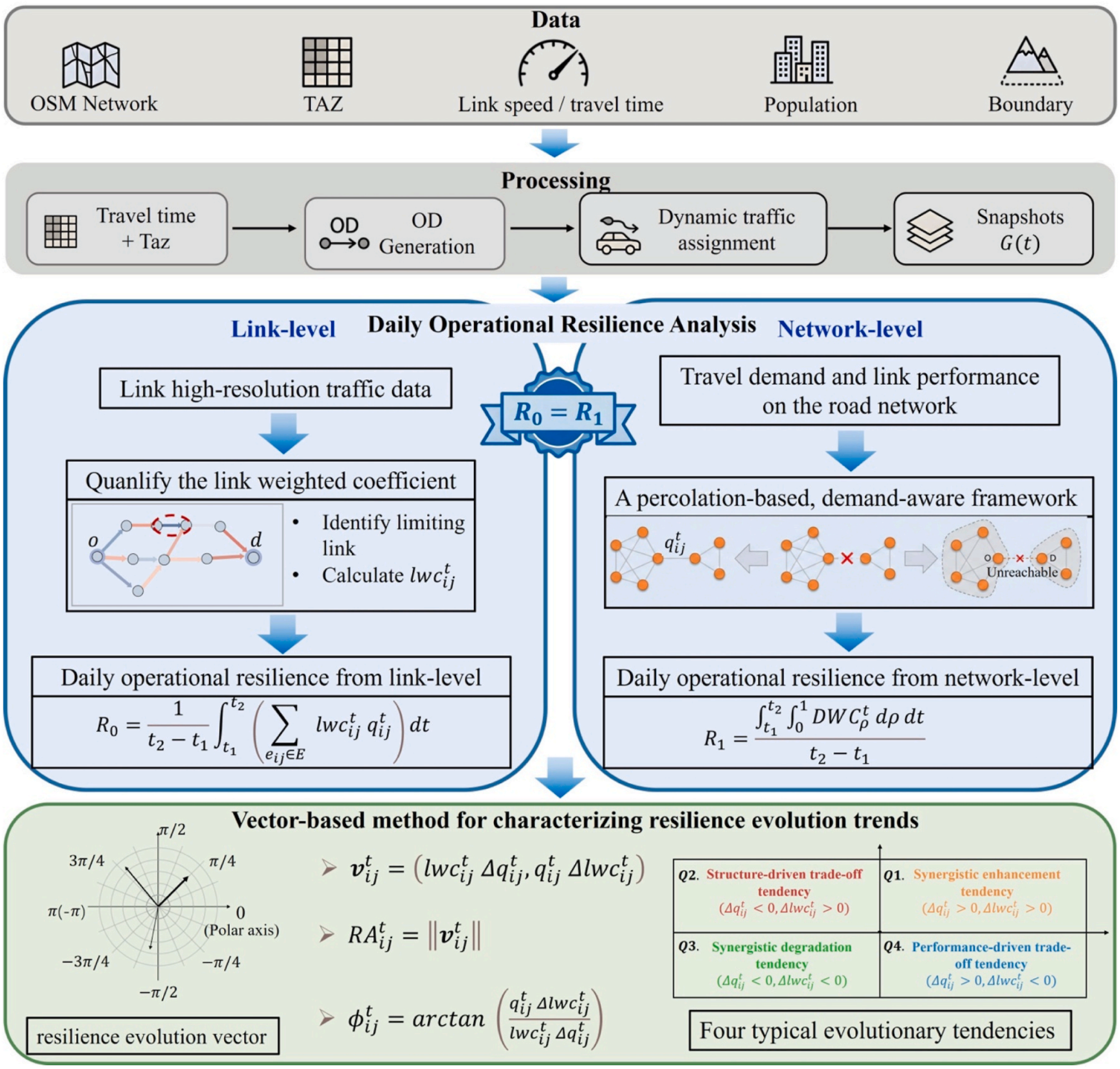



**Fig. 1.** Research framework.

tersections, and links represent road segments between adjacent intersections. The structural topology of the URTS is described by the adjacency matrix $A = [a_{ij}]$:

$$a_{ij} = \begin{cases} 1, \text{if node } i \text{ is connected to node } j \\ 0, \text{otherwise} \end{cases} \tag{3}$$

Travel demand at time interval $t$ is represented by the OD matrix $F^t = [f_{od}^t]$, where $f_{od}^t$ denotes the travel demand from origin node $o$ to destination node $d$ at time interval $t$. Link operational performance is captured by the matrix $Q^t = [q_{ij}^t]$ and $q_{ij}^t$ is the link performance defined in Eq. (1). While $V, E, A$ describe the time-invariant structure, $F^t$ and $Q^t$ characterize time-varying demand and performance, respectively.

### 3.3. Resilience assessment method of URTS

Based on the percolation-based network representation introduced in Sections 3.1-3.2, this study assesses the daily operational resilience of URTS from both network-level and link-level perspectives, which is consistent with the two-level framework in Li et al. (2025). Using link performance $q_{ij}^t \in [0,1]$ and travel demand $f_{od}^t$ as inputs, the framework derives two scale-specific but consistent resilience measures that jointly capture congestion-induced performance degradation and demand-induced structural dependence under daily traffic disturbances. At the link level, we compute the link-weighted coefficient $lwc_{ij}^t$ to summarize demand-weighted OD connectivity dependence on each link (Eq. (4)) and combine $lwc_{ij}^t$ with $q_{ij}^t$ to examine the spatiotemporal distribution of link-level resilience contributions across the network (Eqs. (5) and (6)). At the network level, the framework characterizes dynamic differences in network-wide functional accessibility by constructing demand percolation curves $DWC_\rho^t$ under progressive performance thresholding with reachability updates (Eqs. (7) and (8)). In practice, the network-level measure facilitates cross-period comparisons (e.g., weekday/weekend and peak/off-peak), while the link-level outputs support spatiotemporal localization of resilience contributions. To obtain daily-scale resilience, the measures are aggregated over successive time intervals.

#### 3.3.1. Resilience assessment from a link-level perspective

From a link-level perspective, resilience is computed by aggregating the contributions of individual road segments. For each origin-destination (OD) pair $(o,d)$ at time interval $t$, let $\Psi_{od}^t$ denote the set of all paths connecting $o$ and $d$. For each path in $\Psi_{od}^t$, the link with the minimum $q$ is identified as the bottleneck link. Among all bottleneck links associated with different paths, the one with the highest $q$ is defined as the limiting link $e_{od}^{*,t}$ for the OD pair, assuming link performance $q$ values are unique, so each OD pair has a single limiting link. During the percolation process, as the performance threshold $\rho$ increases, links with performance $q$ below $\rho$ are progressively removed from the network. When $\rho$ reaches the quality level of the limiting link $e^{*,t}_{od}$, the corresponding OD pair loses connectivity (Fig. 2). This indicates the limiting link is the critical element sustaining connectivity for that OD pair. As low-performance links are progressively filtered out, OD pairs that remain reachable switch to alternative feasible paths, so their limiting links may shift to other segments.

Aggregating over all OD pairs at time interval $t$, the link weighted coefficient, following Hamedmoghadam et al. (2021), is defined as

$$lwc^t_{ij} = \sum_{o,d \in V} \mathbf{1}\left(e^{*,t}_{od} = e_{ij}\right) \frac{f^t_{od}}{\mathbf{1}^\top_n F^t \mathbf{1}_n} \tag{4}$$

where $\mathbf{1}(\cdot)$ denotes an indicator function, and $\mathbf{1}^\top_n F^t \mathbf{1}_n = \sum_{o,d} f^t_{od}$ represents the total travel demand in the network. Eq. (4) sums the demand shares of OD pairs whose limiting link is $(i,j)$. Daily operational resilience from the link perspective is then computed as:

$$R_0 = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \left( \sum_{e_{ij} \in E} lwc^t_{ij}\, q^t_{ij} \right) dt \tag{5}$$

Given that traffic data are collected at discrete time intervals, Eq. (5) can be evaluated in discrete form as

$$R_0 = \frac{1}{t_2 - t_1} \sum_{t=t_1}^{t_2} \left( \sum_{e_{ij} \in E} lwc^t_{ij}\, q^t_{ij} \right) \tag{6}$$

### 3.3.2. Resilience assessment from a network-level perspective

From a network-level perspective, daily operational resilience is assessed by examining how OD connectivity degrades as link-level performance deteriorates. For each time interval $t$, the percolation process gradually increases a performance threshold $\rho \in [0,1]$, and all links whose performance satisfies $q^t_{ij} < \rho$ are removed from the network, producing a percolated network under threshold $\rho$. On each percolated network corresponding to threshold $\rho$, OD accessibility is evaluated based on path feasibility. Importantly, the influence of link filtering propagates through connectivity and path feasibility updates. Once a link is filtered out, the set of feasible paths between OD pairs is updated on the modified graph, which may eliminate alternative routes and shift some OD pairs' connectivity dependence to other remaining segments. As $\rho$ increases and more low-performance links are removed, OD reachability degrades endogenously because connectivity is re-evaluated on each updated percolated network, thereby capturing the progressive, congestion-driven loss of functional accessibility under daily operating conditions (illustrated in Fig. 3).

$D^t_\rho$ and $D$ denote the total travel demand associated with reachable OD pairs under threshold $\rho$ and the total travel demand in the network, respectively. The demand weighted coefficient is then defined as

$$DWC^t_\rho = \frac{D^t_\rho}{D} \tag{7}$$

It represents the proportion of total demand associated with OD pairs that remain reachable under performance threshold $\rho$. By varying $\rho$ from 0 to 1, a demand percolation curve $DWC^t_\rho$ is obtained for each time interval $t$, describing the relationship between performance threshold $\rho$ and OD reachability at time $t$. Following established percolation-based resilience formulations (Zhou et al., 2019; Pan et al., 2022), the network-level resilience over a time window $[t_1, t_2]$ is quantified by integrating the demand percolation curves over both performance thresholds and time:

$$R_1 = \frac{\int_{t_1}^{t_2} \int_0^1 DWC^t_\rho \, d\rho dt}{t_2 - t_1} \tag{8}$$

This measure represents the average ability of the network to maintain OD connectivity under recurrent daily traffic disturbances and is consistent with the resilience triangle concept (Adjetey-Bahun et al., 2016).

### 3.3.3. Relationship between network-level and link-level resilience

For each time interval $t$ and performance threshold $\rho$, let $R^t_p = [r^\rho_{od}] \in \{0,1\}^{n \times n}$ denote the OD reachability matrix associated with $G^t_\rho$,

$$r^\rho_{od} = \begin{cases} 1, \text{ifatleastonefeasiblepathexistsbetween} o \text{and} d \text{underthreshold} \rho, \\ 0, \text{otherwise}. \end{cases} \tag{9}$$

The $DWC^t_\rho$ in Eq. (7) can be written as

$$DWC^t_\rho = \frac{D^t_\rho}{D} = \frac{\mathrm{tr}\left(R^t_\rho F^{t\top}\right)}{\mathbf{1}^\top_n F^t \mathbf{1}_n} \tag{10}$$

From the link perspective, recall that for each OD pair $(o,d)$ at time $t$, the limiting link $e^{*,t}_{od}$ is defined as the bottleneck link with the highest performance $q$ among all feasible paths. Under the performance-based percolation process, the connectivity of OD pair $(o,d)$ is lost precisely when the threshold $\rho$ exceeds the performance of its limiting link $q^{*,t}_{od}$. Accordingly, the reachability indicator satisfies

$$r^\rho_{od} = \begin{cases} 1, & \rho \le q^{*,t}_{od}, \\ 0, & \rho > q^{*,t}_{od}, \end{cases} \text{ and } \int_0^1 r^\rho_{od}\, d\rho = q^{*,t}_{od}. \tag{11}$$

Substituting Eq. (11) into Eq. (10) and integrating over $\rho$, it follows that:

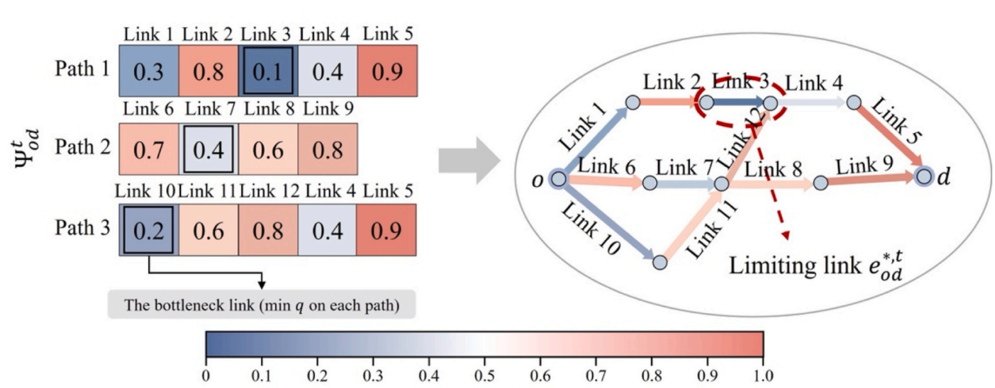



**Fig. 2.** Illustration of bottleneck identification and limiting-link determination for an OD pair.

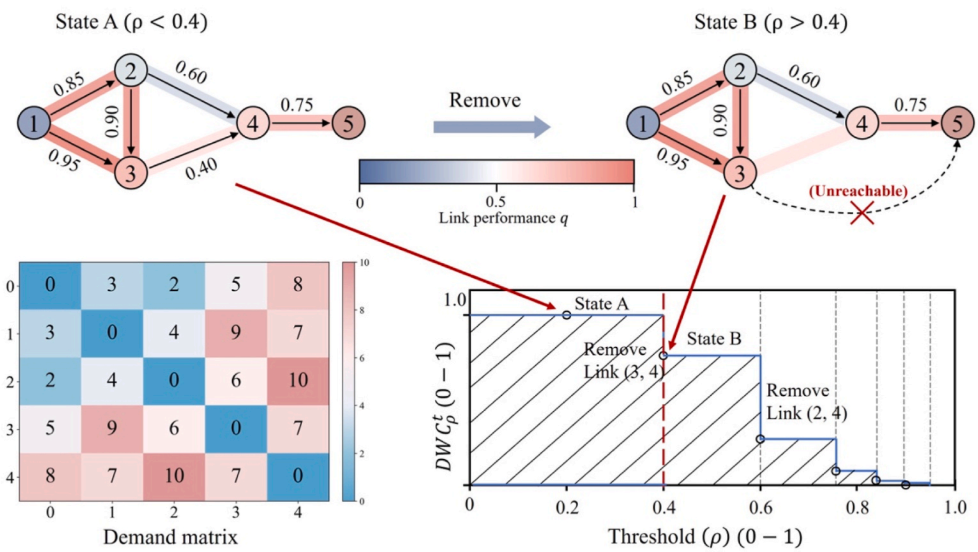



**Fig. 3.** Propagation of performance-based link filtering through OD path feasibility and reachability updates. The "propagation" is not a physical cascade; it arises because removing links with $q_{ij}^t < \rho$ updates feasible path sets and OD reachability on the modified graph, thereby reducing $DWC_\rho^t$. The bottom panel links the network states to the corresponding change on the $DWC_\rho^t$–$\rho$ curve at time interval $t$.

$$\int_0^1 DWC_\rho^t \, d\rho = \frac{1}{\mathbf{1}_n^\top F^t \mathbf{1}_n} \sum_{o,d} f_{od}^t \, q_{od}^{*,t}. \tag{12}$$

On the other hand, by the definition of the $lwc_{ij}^t$ in Eq. (4), aggregating the contribution of all links yields

$$\sum_{e_{ij}\in E} lwc_{ij}^t \, q_{ij}^t = \frac{1}{\mathbf{1}_n^\top F^t \mathbf{1}_n} \sum_{o,d} f_{od}^t \, q_{od}^{*,t}. \tag{13}$$

Combining Eqs. (12) and (13), the following identity holds for each time interval $t$:

$$\int_0^1 DWC_\rho^t \, d\rho = \sum_{e_{ij}\in E} lwc_{ij}^t \, q_{ij}^t \tag{14}$$

Finally, averaging Eq. (14) over a time window $[t_1, t_2]$, the equivalence between the network-level and link-aggregated resilience measures follows directly:

$$R_1 = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \int_0^1 DWC_\rho^t \, d\rho \, dt = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \sum_{e_{ij}\in E} lwc_{ij}^t \, q_{ij}^t \, dt = R_0. \tag{15}$$

This result demonstrates that the proposed network-level resilience measure and the link-level aggregated resilience measure are mathematically equivalent representations of daily operational resilience under the demand-aware percolation framework. Moreover, temporal averaging over $[t_1, t_2]$ enables daily-scale characterization of resilience evolution under recurrent traffic disturbances, rather than relying on a single snapshot.

### 3.4. Vector-based method for characterizing resilience evolution trends

#### 3.4.1. Continuous formulation of resilience evolution

Based on the link-level resilience formulation introduced in Section 3.3.1, the resilience associated with link $(i, j)$ at time $t$ can be expressed as a function of two time-dependent variables: the link performance $q_{ij}(t)$ and the link weighted coefficient $lwc_{ij}(t)$,

$$R_{ij}(t) = R\left(q_{ij}(t), lwc_{ij}(t)\right) \tag{16}$$

Given the multiplicative structure of link-level resilience, this function can be written explicitly as

$$R_{ij}(t) = q_{ij}(t) \, lwc_{ij}(t) \tag{17}$$

Taking the total derivative of $R_{ij}(t)$ with respect to time yields

$$\frac{dR_{ij}(t)}{dt} = \frac{\partial R}{\partial q_{ij}} \frac{dq_{ij}(t)}{dt} + \frac{\partial R}{\partial lwc_{ij}} \frac{d\, lwc_{ij}(t)}{dt} \tag{18}$$

Given Eq. (17), the partial derivatives are obtained directly as

$$\frac{\partial R}{\partial q_{ij}} = lwc_{ij}(t), \frac{\partial R}{\partial lwc_{ij}} = q_{ij}(t) \tag{19}$$

Substituting Eq. (19) into Eq. (18) leads to

$$\frac{dR_{ij}(t)}{dt} = lwc_{ij}(t) \frac{dq_{ij}(t)}{dt} + q_{ij}(t) \frac{d\, lwc_{ij}(t)}{dt} \tag{20}$$

Eq. (20) shows that the temporal evolution of link-level resilience is governed by two first-order mechanisms: variations in traffic operating performance and changes in structural dependence captured by the demand-weighted importance of the link. Because traffic states are observed at discrete intervals, the continuous-time derivatives in Eq. (20) are approximated using first-order finite differences:

$$\frac{dq_{ij}(t)}{dt} \approx q_{ij}^{t+1} - q_{ij}^t, \frac{d\, lwc_{ij}(t)}{dt} \approx lwc_{ij}^{t+1} - lwc_{ij}^t \tag{21}$$

Substituting Eq. (21) into Eq. (20), the discrete variation of link-level resilience between two consecutive time intervals is obtained as

$$\Delta R_{ij}^t = R_{ij}^{t+1} - R_{ij}^t \approx lwc_{ij}^t \, \Delta q_{ij}^t + q_{ij}^t \, \Delta lwc_{ij}^t \tag{22}$$

where

$$\Delta q_{ij}^t = q_{ij}^{t+1} - q_{ij}^t, \Delta lwc_{ij}^t = lwc_{ij}^{t+1} - lwc_{ij}^t \tag{23}$$

#### 3.4.2. Vector-based characterization of resilience evolution

While the scalar quantity $\Delta R_{ij}^t$ indicates whether link-level resilience improves, it does not reveal the underlying mechanisms driving this change. To address this limitation, a vector-based representation of resilience evolution is introduced. The method provides a mechanism-oriented interpretation of the direction and intensity of link-level resilience evolution.

For each link $(i, j)$ at time interval $t$, the resilience evolution vector is defined as

$$\mathbf{v}_{ij}^t = \left(lwc_{ij}^t\,\Delta q_{ij}^t, q_{ij}^t\,\Delta lwc_{ij}^t\right) \tag{24}$$

The two components of $\mathbf{v}_{ij}^t$ respectively capture the contributions of traffic operating performance variation and structural dependence reconfiguration to resilience evolution. Eq. (22) provides the corresponding scalar resilience variation. To further characterize trends, the vector $\mathbf{v}_{ij}^t$ is expressed in polar form. A schematic illustration of the polar-coordinate representation of $\mathbf{v}_{ij}^t$ is shown in Fig. 4. The magnitude of resilience evolution is defined as

$$RA_{ij}^t = \left\|\mathbf{v}_{ij}^t\right\| = \sqrt{\left(lwc_{ij}^t\,\Delta q_{ij}^t\right)^2 + \left(q_{ij}^t\,\Delta lwc_{ij}^t\right)^2} \tag{25}$$

which reflects the intensity of resilience change. The direction of resilience evolution is defined as

$$\phi_{ij}^t = arctan\left(\frac{q_{ij}^t\,\Delta lwc_{ij}^t}{lwc_{ij}^t\,\Delta q_{ij}^t}\right) \tag{26}$$

indicating the dominant mechanism driving resilience evolution, ranging from performance-driven to structure-driven dynamics. Together, $\Delta R_{ij}^t$, $R\,A_{ij}^t$, and $\phi_{ij}^t$ provide a unified description of whether resilience changes, how strongly it changes, and through which mechanism it evolves over time.

*3.4.3. Classification of resilience evolutionary tendencies*

Building on the proposed vector-based formulation, this section introduces a mechanism-oriented classification of resilience evolutionary tendencies to characterize typical patterns of link-level resilience evolution under daily traffic disturbances. The classification is based on the sign combination of $\left(\Delta q_{ij}^t, \Delta lwc_{ij}^t\right)$, while the net outcome is determined by $\Delta R_{ij}^t$. Four typical evolutionary tendencies are identified and are shown in Fig. 5.

*3.4.3.1. Synergistic enhancement tendency*

$\Delta q_{ij}^t > 0, \Delta lwc_{ij}^t > 0(27)$

Both traffic operating performance and structural dependence improve simultaneously, leading to an unambiguously positive resilience variation:

$\Delta R_{ij}^t > 0(28)$

*3.4.3.2. Performance-driven trade-off tendency*

$\Delta q_{ij}^t > 0, \Delta lwc_{ij}^t < 0(29)$

Traffic operating performance improves while structural dependence decreases. The two first-order components exert opposing effects, and the sign of

$$\Delta R_{ij}^t \approx lwc_{ij}^t\,\Delta q_{ij}^t + q_{ij}^t\,\Delta lwc_{ij}^t \tag{30}$$

depends on their relative magnitudes.

*3.4.3.3. Structure-driven trade-off tendency*

$\Delta q_{ij}^t < 0, \Delta lwc_{ij}^t > 0(31)$

Traffic operating performance deteriorates while structural dependence intensifies. The net outcome depends on the balance between the two components.

*3.4.3.4. Synergistic degradation tendency*

$\Delta q_{ij}^t < 0, \Delta lwc_{ij}^t < 0(32)$

Both components deteriorate simultaneously, resulting in an unambiguously negative resilience variation:

$\Delta R_{ij}^t < 0.(33)$

The zero-threshold used to classify the signs of $\Delta q_{ij}^t$ and $\Delta \mathrm{lwc}_{ij}^t$ follows directly from the first-order formulation of resilience evolution. Although measurement noise may affect the magnitudes of these variations, the classification itself is not magnitude-based. A sensitivity analysis examining the robustness of the proposed classification under perturbations in link-average speed is provided in *Appendix B*.

## 4. Case study

This section applies the proposed framework to the road network in Nanjing's central districts to demonstrate its applicability to a large-scale real-world network (Fig. 6). In this case study, the proposed framework is implemented using time-varying network snapshots constructed from 15-min link performance $q_{ij}^t$ and OD demand $f_{od}^t$. These inputs yield two consistent sets of outputs: at the link level, limiting-link identification is summarized into the link weighted coefficient $lwc_{ij}^t$, which is aggregated (15-min to hourly) to describe the spatiotemporal distribution and temporal shifts of link criticality (Section 4.2); at the network level, demand percolation curves $DWC_\rho^t$ and their temporal aggregation yield resilience measures used for benchmarking daily operational conditions across day types, peak periods, and seasons, with additional robustness checks against alternative metrics and demand-metric consistency (Section 4.3). Based on the above two-level assessment, Section 4.4 proceeds to characterize link-level resilience evolution by jointly considering variations in operating performance $q_{ij}^t$ and demand-weighted structural dependence $lwc_{ij}^t$.

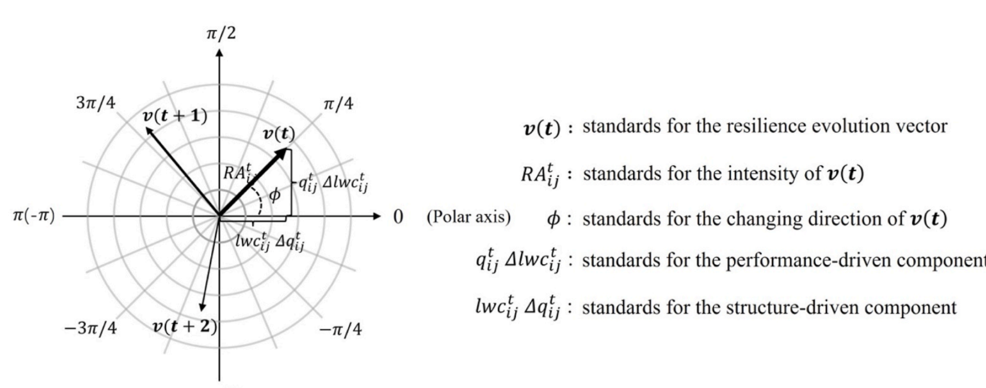



**Fig. 4.** Time-varying resilience evolution vector on polar coordinate.

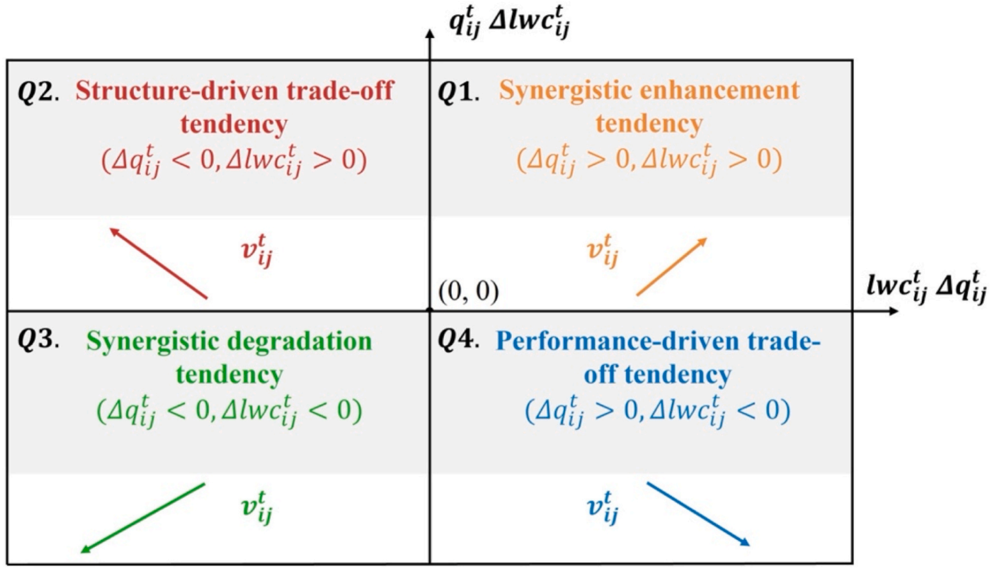



**Fig. 5.** The meanings of the four typical evolutionary tendencies.

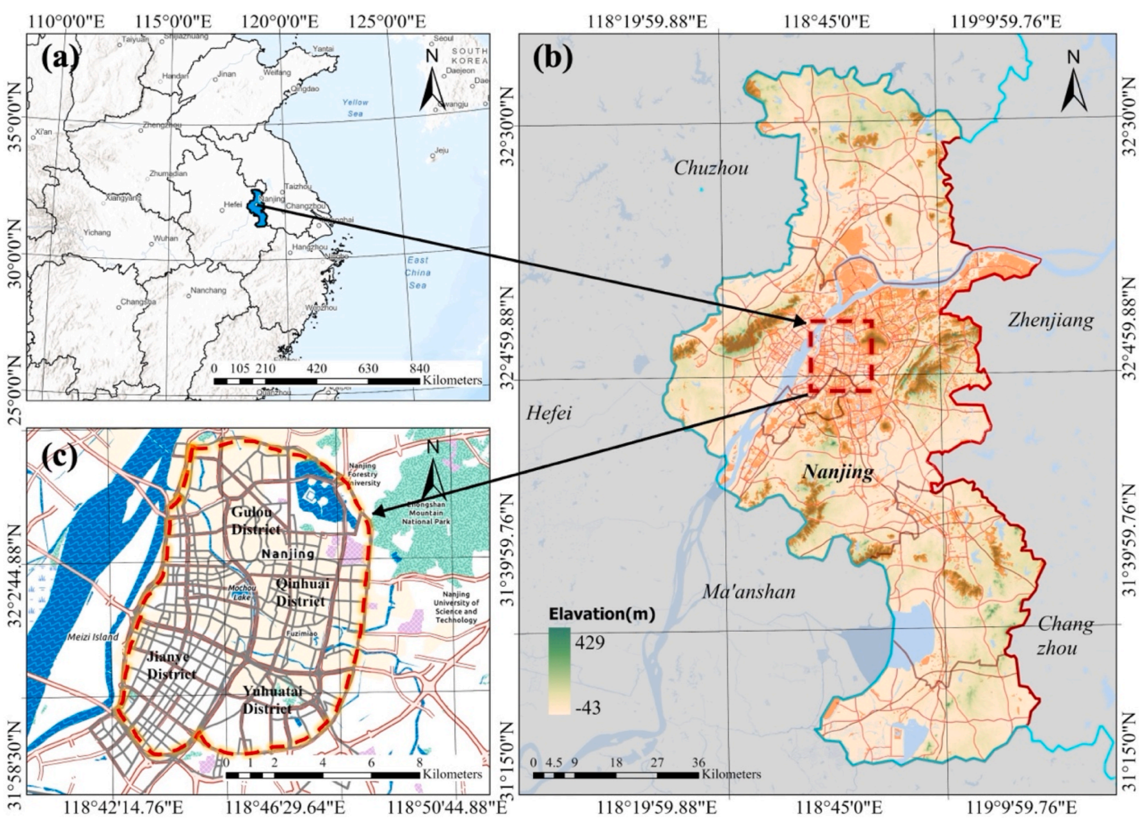



**Fig. 6.** Location of the study area.

## 4.1. Data collected

The data used in this case study are obtained from the Nanjing Statistical Yearbook, OpenStreetMap, Geospatial Data Cloud, and Gaode Map Web Service APIs. The data types and sources are summarized in Table 2. Due to differences in spatial accuracy among population density data, age structure and DEM data, this research conducts resampling analysis and standardizes their spatial accuracy to 100m.

To capture the temporal variability of daily traffic operations, origin-destination (OD) demand matrices are generated at a 15-min interval for two representative periods covering May 25-June 8 and Dec 22-Dec 28, 2025. Population density and age structure data are spatially aggregated to traffic analysis zones (TAZs) to estimate trip productions and attractions. A gravity model is adopted to estimate inter-zonal trip distribution, with its impedance function calibrated using observed travel times between TAZs obtained from the Gaode Map Route Planning API. These travel time data are continuously collected at 15-min intervals over the study period, ensuring that the generated OD matrices reflect realistic and time-varying travel impedance under daily operating conditions. Based on the 15-min OD demand matrices, dynamic traffic assignment is

**Table 2**
Sources of regional city network construction data.

| Type | Data | Sources | Resolution |
|---|---|---|---|
| The URTS topology data | Network topology (URTS) | Open Street Map (https://www.openstreetmap.org/) | Vector |
| Traffic demand data | Population density data | 2024 Nanjing Statistical Yearbook | 100m |
| | Age structure | 2024 Nanjing Statistical Yearbook | 100m |
| | DEM | Geospatial Data Cloud (https://www.gscloud.cn/) | 30m |
| | Administrative boundary | Geospatial Data Cloud (https://www.gscloud.cn/) | Vector |
| | Traffic analysis zone | Open Street Map (https://www.openstreetmap.org/) | Vector |
| | Travel time between two TAZs | The Gaode Map Route Planning API (https://lbs.amap.com/api/webservice/guide/api/direction) | 22 days |
| Link quality data | Road segments operation status data | The Gaode Map Traffic Operation States API (https://lbs.amap.com/api/webservice/guide/api-advanced/traffic-situation-inquiry) | 22 days |

performed to obtain time-dependent link-level traffic states, which serve as the basis for constructing network snapshots. The simulation is executed as a continuous process rather than independent 15-min runs: demand is loaded at a 15-min resolution, while congestion formation, queue propagation, and unfinished trips persist across intervals, thereby affecting subsequent link travel times and performance states. To evaluate the representativeness of the simulated network states, the traffic assignment results are compared with observed link performance data obtained from the Gaode Map Traffic Operation States API. Since the Gaode traffic operation data provide reliable speed information primarily for major urban roads, the comparison focuses on the corresponding road segments included in the simplified URTS (Fig. 7). The trend comparison yields an $R^2$ of 81.26%, which falls within the 65%-90% accuracy range commonly accepted and reported in previous traffic simulation studies (Laña et al., 2019; Li et al., 2020).

### 4.2. Network construction and the link weighted coefficient computation

The static network topology, together with the time-dependent traffic demand and link performance data processed for May 25-June 8 and Dec 22-Dec 28, 2025, is used to construct a time-varying URTS network for Nanjing, as described in Section 3.2. Since both the generated OD demand and the observed link performance data are available at a 15-min resolution, the URTS is represented as a sequence of 15-min network snapshots.

The 15-min link weighted coefficients are aggregated to an hourly scale by averaging across consecutive snapshots. Table 3 reports the top 10 links with the highest *lwc* values for three typical periods (8:00-9:00, 12:00-13:00, and 18:00-19:00) on 29 May. The *lwc* values quantify the demand-weighted structural contribution (criticality) of each link during the specified period. The results show pronounced temporal heterogeneity: the most critical links shift across periods, such as Inner Ring East Road in the morning, South Lake Road at noon, and the Nanjing Yangtze River Bridge in the evening. Meanwhile, several links, such as Lingyuan Road, remain consistently influential, indicating persistent structural importance under daily traffic disturbances. These findings provide a quantitative basis for subsequent resilience assessment and evolutionary classification.

### 4.3. Resilience assessment analysis of URTS

#### 4.3.1. Fluctuating patterns of daily operational resilience in URTS

Understanding temporal variability is essential for capturing how URTS respond to everyday operational pressures. Long-term averages may mask short-term fluctuations across weekdays/weekends, peak periods, holidays, and seasons. This section therefore examines daily resilience patterns in Nanjing's URTS across weekly cycles, aggregated day types, peak periods, and seasonal contexts to reveal systematic temporal regularities and anomalies under routine traffic conditions.

Fig. 8 presents the weekly pattern and day-type comparison of mean resilience in the URTS. At the weekly scale (Fig. 8(a)), resilience exhibits a gradual upward trend from Monday (0.742) to Sunday (0.783), forming a clear “Monday-low to Sunday-high” pattern. This pattern reflects systematic differences in resilience levels across the week under daily operating conditions. A similar tendency is observed in the aggregated day-type classification (Fig. 8(b)): Weekdays show the lowest mean resilience (0.749), while pure weekends exhibit the highest mean resilience (0.777), with holiday-weekends showing intermediate values (0.759). To examine whether the observed weekday-weekend contrast is statistically meaningful, a Mann-Whitney $U$ test was conducted based on daily aggregated resilience samples, which is consistent with Van Essen et al. (2019). The results indicate the difference between

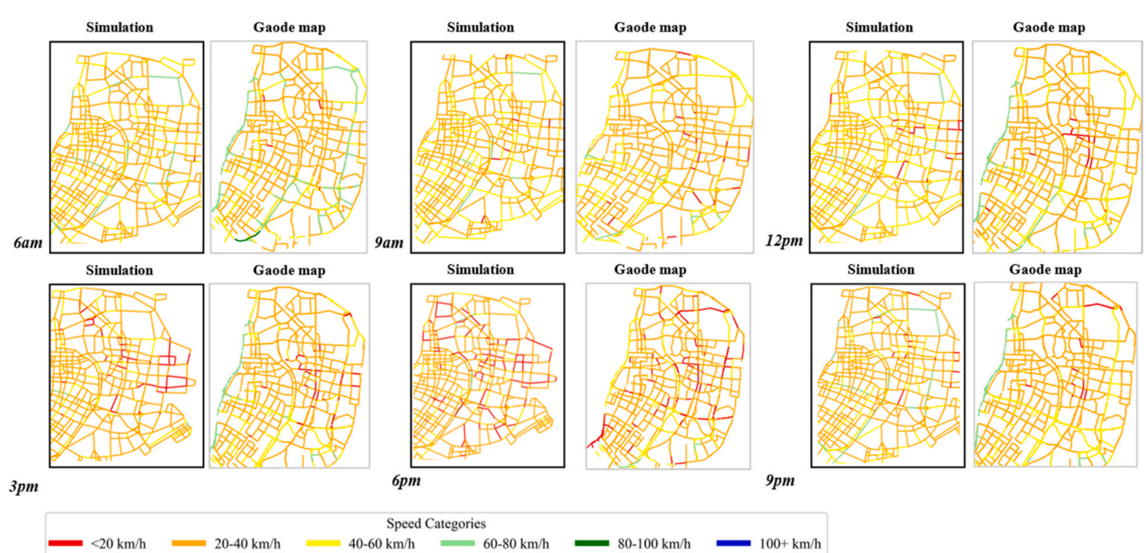



**Fig. 7.** Comparison between simulation result and Gaode Map data.

**Table 3**
The hourly ranking of the weighted coefficient of the URTS on 29 May.

| Rank | Morning Peak (8:00-9:00) | | Noon (12:00-13:00) | | Evening Peak (18:00-19:00) | |
|---|---|---|---|---|---|---|
| | Link | *lwc* | Link | *lwc* | Link | *lwc* |
| 1 | Inner Ring East Road (Xi'anmen Gate) | 0.2931 | South Lake Road | 0.1343 | Nanjing Yangtze River Bridge | 0.2143 |
| 2 | Lingyuan Road | 0.0908 | Yudao Street | 0.1056 | Lingyuan Road | 0.0969 |
| 3 | Jiangshan Avenue | 0.0681 | East Zhongshan Road (Xinjiekou) | 0.0761 | East Zhongshan Road | 0.0577 |
| 4 | Inner Ring North Road (Gupinggang) | 0.0347 | Lingyuan Road | 0.0557 | Caochangmen Avenue | 0.0374 |
| 5 | Kazimen Avenue | 0.0331 | Huangpu Road | 0.0434 | Guangzhou Road | 0.0371 |
| 6 | East Zhongshan Road | 0.0323 | Guangzhou Road | 0.0430 | Dashiqiao Street | 0.0368 |
| 7 | Ruijin Road | 0.0306 | Jinxiang River Road | 0.0419 | Ruijin Road | 0.0313 |
| 8 | Inner Ring East Road (Shuangqiaomen Interchange) | 0.0275 | Inner Ring West Road (Nanjing Yangtze River Bridge) | 0.0416 | Jianning Road | 0.0301 |
| 9 | Huangpu Road | 0.0254 | Qingliangmen Avenue | 0.0345 | Huaqiao Road | 0.0271 |
| 10 | Zhujiang Road | 0.0246 | Zhujiang Road | 0.0313 | Qingxi Road | 0.0271 |

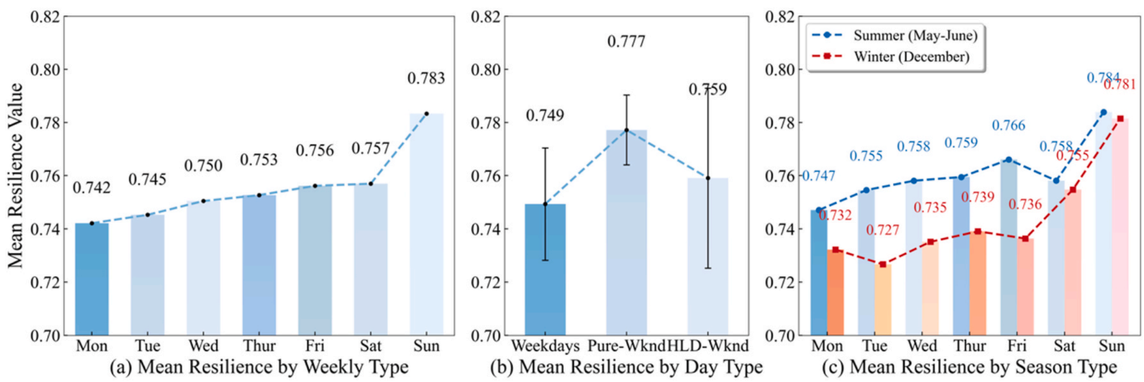



**Fig. 8.** Weekly Pattern and Day-Type Comparison of Mean Resilience in the URTS. Error bars represent the standard deviation of daily aggregated resilience values.

weekdays and pure weekends is statistically significant ($U = 0.0, p = 0.009$), confirming that resilience levels under pure weekend conditions are systematically higher than those observed during weekdays. In addition, holiday-weekends are characterized by noticeably wider error bars than weekdays and pure weekends, indicating larger inter-day variability in resilience values during holiday periods. Fig. 8(c) compares daily resilience patterns between summer and winter. Overall, winter resilience levels are consistently lower than those in summer across most weekdays, reflecting the potentially suppressing effect of adverse seasonal operating conditions. Notably, the pronounced dip observed on Tuesday in winter is associated with rainfall on that day, indicating that weather-related disturbances can further amplify weekday vulnerability beyond seasonal effects alone.

At the daily peak-hour level (Fig. 9), resilience exhibits pronounced fluctuations across the 14-day observation window. For all observed days, morning peak resilience values (7:00-10:00) are consistently lower than evening peak values (17:00-20:00), as illustrated by the contrast on May 26 (0.625 vs. 0.635) and subsequent weekdays. This peak-hour

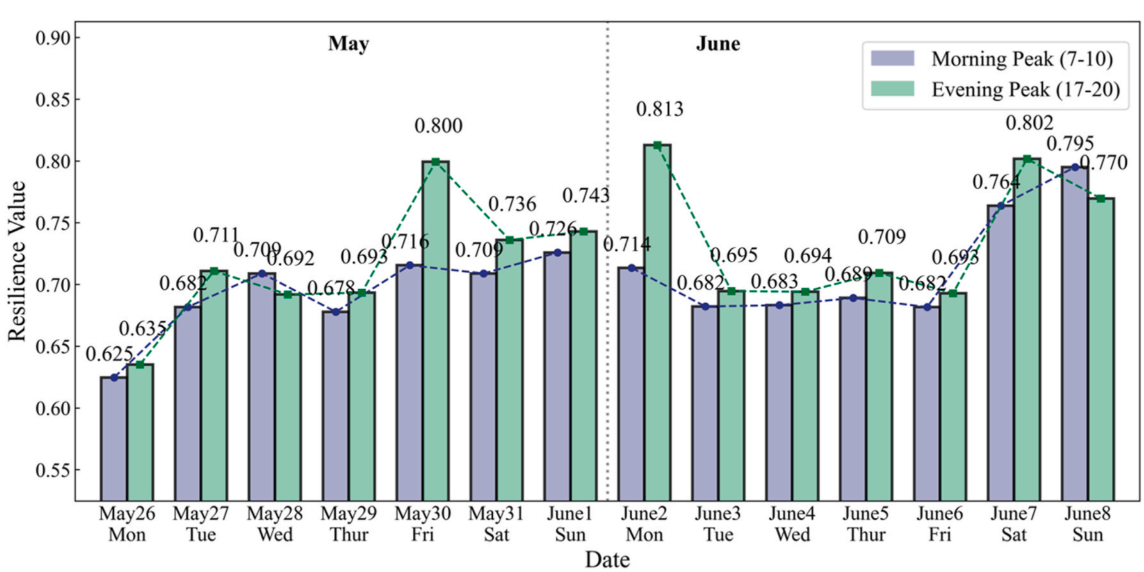



**Fig. 9.** Daily peak-hour resilience comparison across 14 days.

asymmetry persists throughout the study period. Distinct short-term anomalies are observed around the Dragon Boat Festival period (May 31-June 2). Elevated resilience values occur on the last workday before the holiday (Friday, May 30), with resilience reaching 0.716 during the morning peak and 0.800 during the evening peak. Similarly, the final day of the holiday (Monday, June 2) records high resilience levels of 0.714 (morning) and 0.813 (evening), representing the highest evening-peak value observed during the study period. Following the holiday, resilience declines over the subsequent weekdays, with morning and evening peak values ranging between 0.682 and 0.695, indicating a clear post-holiday adjustment in daily operational conditions.

Overall, daily operational resilience exhibits clear temporal variability across weekly cycles, day types, peak periods, and seasons, including a significant weekday-weekend contrast ($U = 0.0, p = 0.009$), persistent morning-evening peak asymmetry, holiday-related anomalies, and lower winter levels with rainfall-amplified dips.

#### 4.3.2. Stability and robustness analysis of resilience assessment methods

Fig. 10(a) illustrates the temporal evolution of the resilience metric $R$ between 6:00 and 22:00 for weekdays, pure weekends, and holiday-weekends. Across all day types, resilience exhibits a coherent and repeatable diurnal pattern, with higher levels during off-peak periods and pronounced declines during peak-demand hours. A clear weekday-weekend contrast is observed: weekdays show deeper peak-hour reductions, whereas pure weekends maintain higher resilience and milder peak-hour declines. Holiday-weekends exhibit an even lower minimum than pure weekends, indicating amplified intra-day fluctuations under holiday-related travel demand. For weekdays, the twin-peak commuting rhythm remains evident, while weekend travel produces a flatter and less volatile profile. Fig. 10(b) presents the temporal variation of the size of the largest connected component. Compared with the resilience metric $R$, the *LCC* size fluctuates irregularly throughout the day and does not exhibit a clear or consistent temporal pattern across different day types, limiting its ability to characterize intra-day resilience dynamics under daily operating conditions.

As shown in Fig. 11, the temporally aggregated resilience measure derived from the proposed framework is benchmarked against weighted average path length, weighted global efficiency, and weighted average betweenness centrality. On May 29 (weekday), all four measures reproduce the expected twin-peak pattern of commuting flows. On May 31 (weekend), when rush-hour peaks weaken and demand becomes more evenly distributed, only the temporally aggregated resilience measure captures this shift, yielding a single, broad resilience dip (Fig. 11(a)). In contrast, average path length and betweenness also capture weekend flattening but exhibit higher variability in response to minor demand swings (Fig. 11(b) and (d)), while global efficiency does not exhibit a stable weekend pattern and shows larger fluctuations (Fig. 11(c)). Collectively, these comparisons indicate that the proposed framework is more consistent in reflecting daily mobility rhythms and more robust in distinguishing between weekday commuting and weekend discretionary travel.

Fig. 12(a) shows the temporal evolution of travel demand, with pronounced morning and evening peaks on weekdays and a flatter, more evenly distributed profile on weekends. These day-type demand profiles correspond to the temporal variations captured by the temporally aggregated resilience measure, indicating demand-resilience consistency under daily operating conditions. To quantify this relationship, Fig. 12(b) and Table 4 report Spearman correlation results between travel demand and different resilience-related metrics. On the weekday, all metrics exhibit statistically significant negative correlations with demand and show relatively strong demand-metric consistency. On the pure weekend, the framework-derived resilience measure exhibits the strongest demand-metric consistency ($p < 0.001$), whereas weighted global efficiency becomes non-significant. Some benchmark metrics are more sensitive to minor demand variations and yield less stable pure-weekend patterns, whereas the proposed measure remains stable by combining demand-aware percolation with temporal aggregation. This cross-day-type stability supports its use for daily operational resilience evaluation and mechanism-aware traffic management.

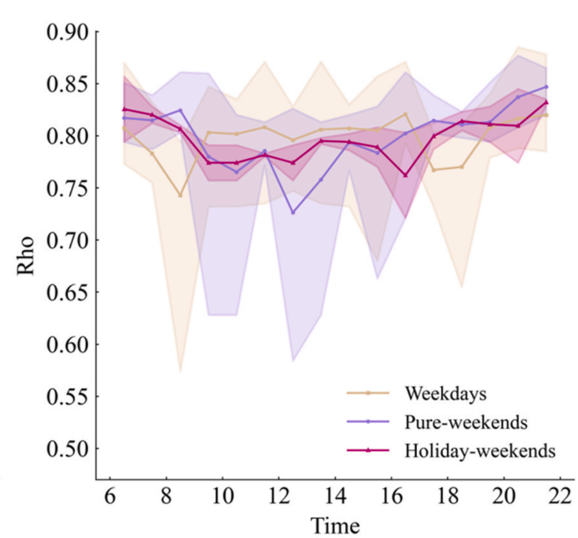
### 4.4. Resilience trend and evolutionary mechanism analysis

#### 4.4.1. Maps representing the global evolution pattern of the time-varying resilience index

Fig. 13 is a zoomed-in detail showing the time-varying resilience evolution vector $\boldsymbol{v}^t$, extending the analysis from resilience levels to resilience evolution trends. It enables a systematic examination of how link-level resilience evolves over time and offers a mechanism-oriented interpretation of resilience dynamics. As mentioned in Section 3.4.2, the arrows represent the direction of resilience evolution, the length of the arrows denote the relative value $RA^t$, and colors denote different time intervals.

Fig. 13 also shows the resilience evolution vector maps at three different times: 8:30, 12:30, and 18:30. According to Fig. 11(a), the

(a) Resilience of the Nanjing road network



(b) Size of largest connected component

**Fig. 10.** Temporal evolution of resilience $R$ and the largest connected component (LCC) size between 6:00 and 22:00 for weekdays, pure weekends, and holiday-weekends.

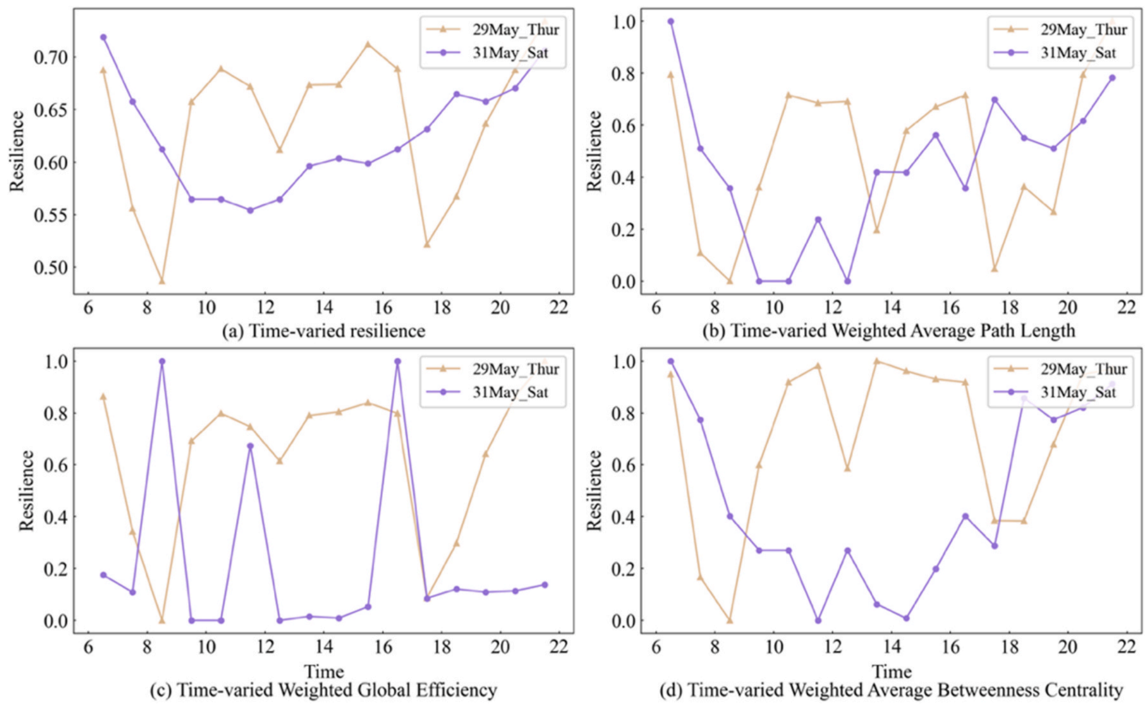



**Fig. 11.** Comparison of the proposed framework and three performance-based resilience assessment methods in urban traffic networks.

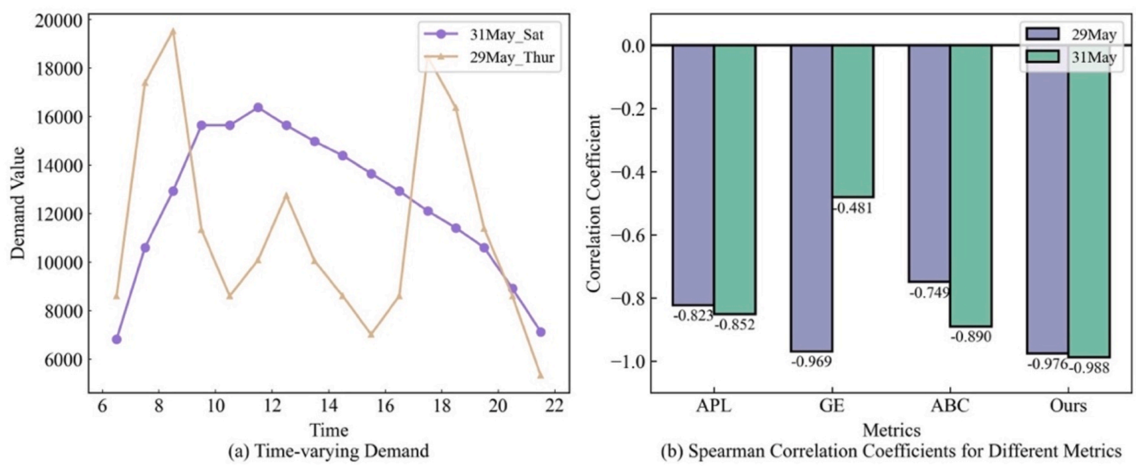



**Fig. 12.** Comparison of daily evolution of $R$ with network travel demand and Spearman correlation.

network-level resilience reaches its lowest value between 8:00 and 9:00, reflecting peak-period operating stress. This temporal low point is associated with the spatial patterns observed in Fig. 14(a), where numerous red arrows are concentrated along major arterials and ring roads.

Notably, many links have exhibited a high-pressure recovery trend that operating performance improves while the network remains under peak-hour demand pressure. At midday (12:00-13:00), Fig. 14(b) exhibits scattered and shorter blue arrows, reflecting localized adjustments with limited system-wide coherence. This suggests a more stable regime in which travel demand is dispersed and overall resilience remains high, with only minor perturbations observed at local bottlenecks. During the evening peak (18:00-19:00), Fig. 14(c) shows that green arrows are concentrated along key cross-river corridors and trunk roads, highlighting that evening commuting imposes directional stress on specific corridors rather than generating uniformly low resilience across the network. Compared with the morning peak, the evening recovery trend is more gradual, reflecting greater temporal dispersion and route flexibility.

#### 4.4.2. Resilience evolutionary tendency map

To identify typical link-level evolution patterns under daily traffic disturbances, resilience evolutionary tendencies are classified by the joint signs of the first-order variations in link performance ($\Delta q$) and structural dependence ($\Delta lwc$), based on the vector-based formulation introduced in Section 3.4.2. This scheme characterizes evolution trend rather than resilience level. Fig. 15 illustrates the spatial distribution of these evolutionary tendencies over time. Three recurring diurnal patterns are observed:

(1) Morning peak (7:30-9:30): synergistic degradation and structure-driven trade-off dominate along major arterials and central corridors, indicating either concurrent deterioration or intensified structural dependence despite performance degradation. This

**Table 4**
Statistical significance of correlations between different metrics.

| Day type | Metric | Spearman correlation | p-value | Significance |
|---|---|---|---|---|
| Weekday (29 May) | Average path length | −0.823 | $9.01 \times 10^{-5}$ | ✓ |
| | Global efficiency | −0.979 | $4.06 \times 10^{-11}$ | ✓ |
| | Average betweenness centrality | −0.749 | $8.35 \times 10^{-4}$ | ✓ |
| | **This study** | −0.976 | $1.03 \times 10^{-10}$ | ✓ |
| Weekend (31 May) | Average path length | −0.852 | $2.84 \times 10^{-5}$ | ✓ |
| | Global efficiency | −0.481 | $5.94 \times 10^{-2}$ | × |
| | Average betweenness centrality | −0.890 | $3.84 \times 10^{-6}$ | ✓ |
| | **This study** | −0.988 | $8.63 \times 10^{-13}$ | ✓ |

spatial pattern reflects pronounced resilience deterioration during concentrated commuting periods.

(2) Midday (10:30-15:30) and late evening (20:30-21:30): performance-driven trade-off and synergistic enhancement become more prevalent, with fewer large degradation clusters, indicating relatively stable evolution with localized fluctuations.
(3) Evening peak (17:30-19:30): degradation-related tendencies re-emerge on critical corridors (cross-river bridges and radial links), but the pattern is more heterogeneous than in the morning, concentrating on specific corridors rather than across the network.

To further summarize the temporal distribution of resilience evolutionary tendencies, Fig. 16 presents the proportion of road segments associated with each tendency across different time intervals. During the morning peak, the proportion of synergistic enhancement increases despite the overall decline in network-level resilience, indicating that while the system remains under stress, a substantial number of links exhibit positive resilience evolution trends. During off-peak periods, the distribution across the four tendencies becomes more balanced,

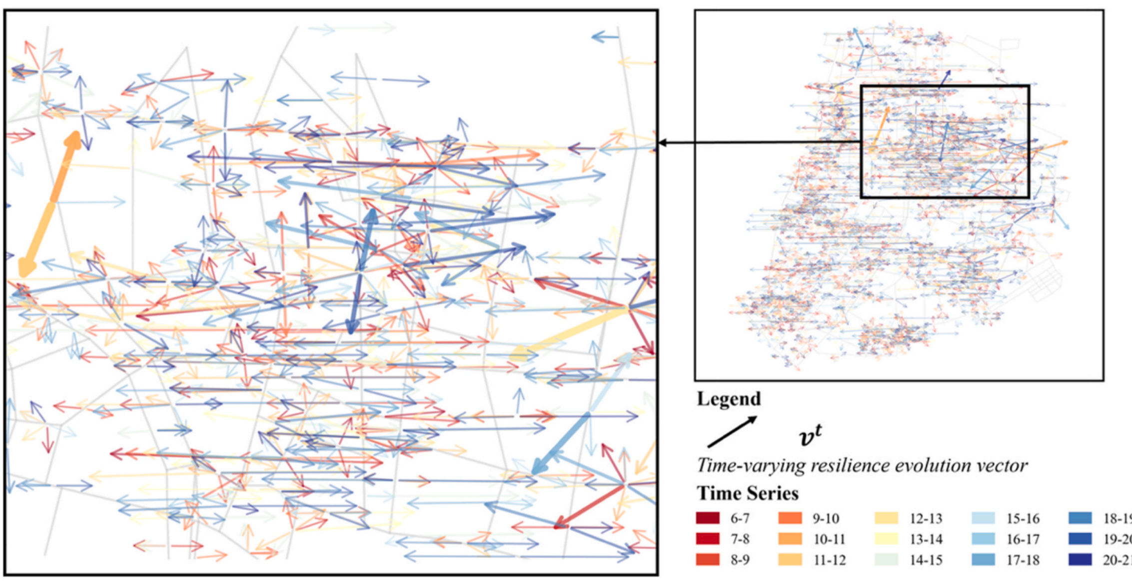



**Fig. 13.** Resilience evolution vectors in the study area on May 29, 2025.

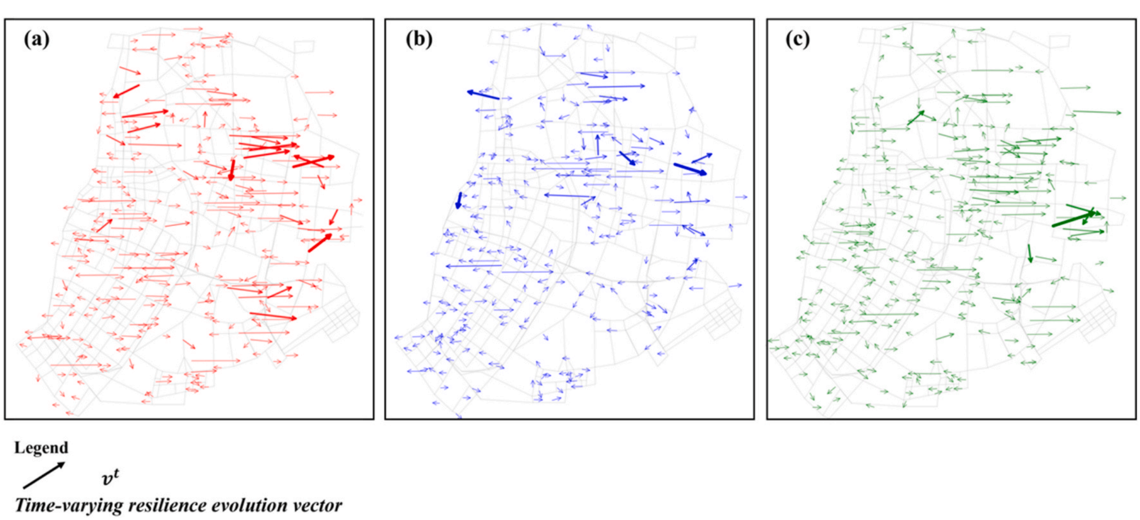



**Fig. 14.** Resilience evolution vectors across time periods (a) 8am to 9am (b) 12pm to 13pm (c) 18pm to 19pm on May 29, 2025.

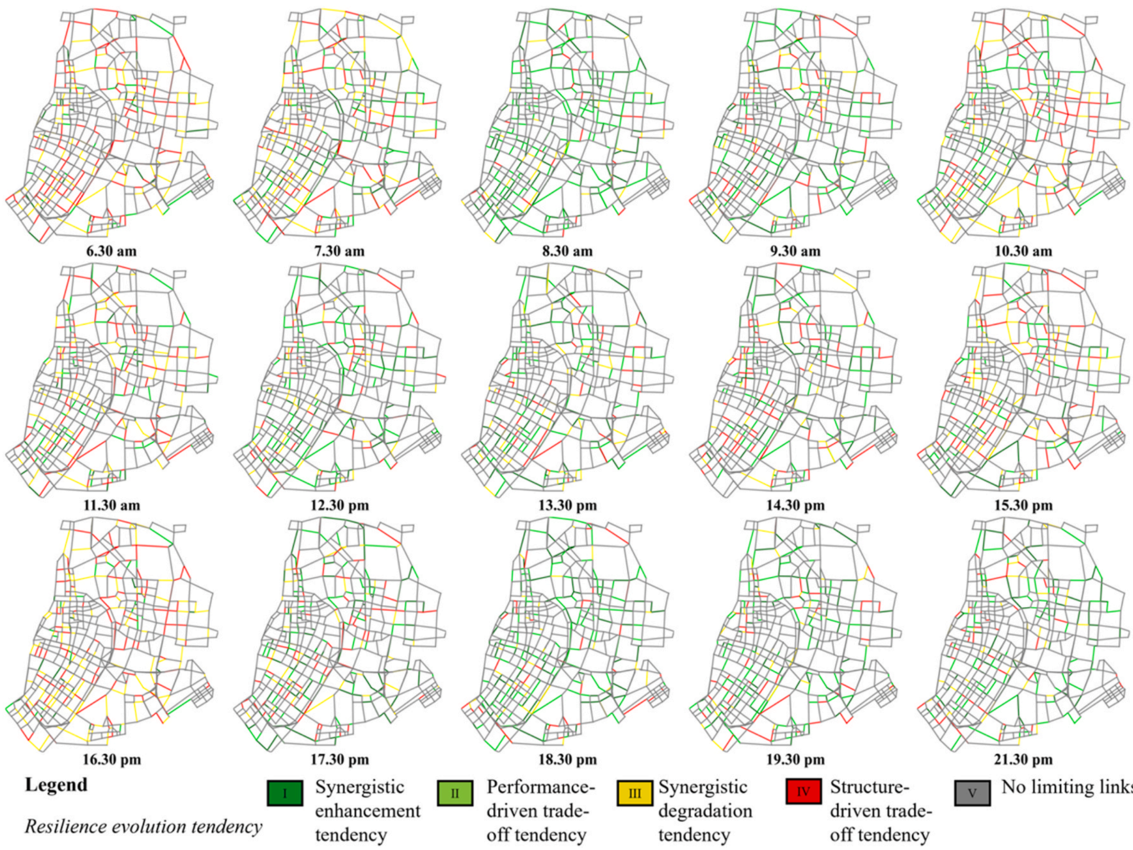



**Fig. 15.** Time-varying resilience evolutionary tendency map.

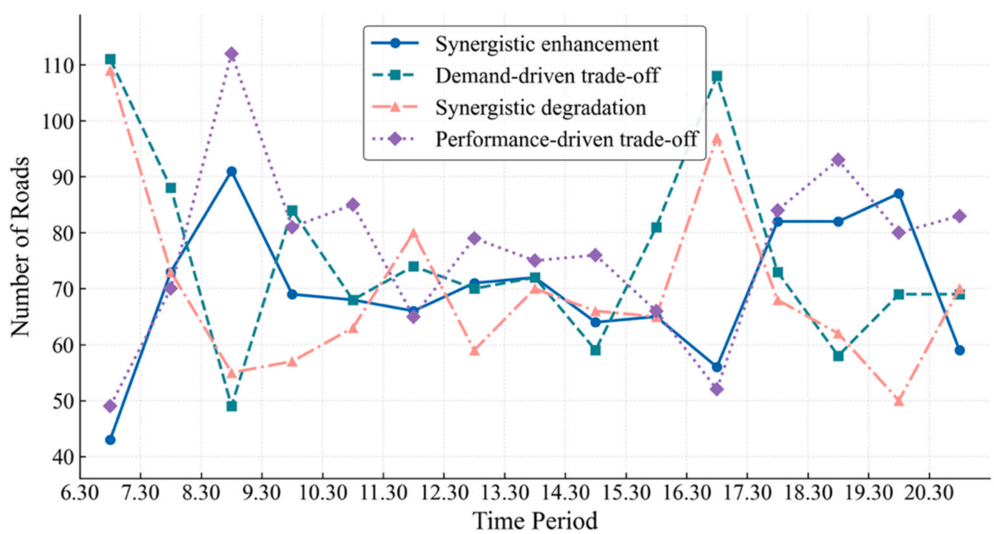



**Fig. 16.** Resilience evolutionary tendency in road segments at different time intervals.

reflecting relatively steady resilience evolution at the network scale. A similar pattern is observed during the evening peak, although transitions among tendencies occur more gradually than in the morning.

Overall, the tendency maps reveal (i) clear morning-evening asymmetry and (ii) concurrent mechanisms across links at any time. By decoupling evolution trends from absolute resilience levels, the results provide a complementary, mechanism-oriented view of daily operational resilience; Fig. 16 quantifies this view via the temporal composition of the four tendencies.

## 5. Discussion and policy implications

### 5.1. Interpreting daily resilience dynamics under routine traffic disturbances

Existing studies have demonstrated that transportation system resilience is not determined solely by static network structure, but exhibits pronounced temporal variability as operating conditions and demand levels change (Liu et al., 2025). Extending this perspective to daily operations, this study shows that resilience varies systematically across day types, peak periods, and seasonal conditions, reflecting recurrent travel routines and demand redistribution. In particular, the weekday-weekend contrast underscores how demand regularity and activity scheduling shape daily operational resilience.

Mechanistically, routine commuting periods impose synchronized and directional pressure on critical corridors, whereas more flexible travel patterns allow demand to be redistributed across time and space. This mechanism is conceptually aligned with Zhu et al. (2025), who reported pronounced resilience degradation and shifting network bottlenecks during synchronized morning-peak operations. As a result, resilience should be understood as a condition-dependent system property that reflects how demand is organized and accommodated, rather than as a static attribute of network connectivity. The observed morning-evening asymmetry further implies time-specific management needs. The stronger synchronization and directional concentration of morning commuting flows suggest that resilience degradation during this period is less amenable to spontaneous demand redistribution. Consequently, morning peak management should prioritize proactive measures, such as demand suppression, staggered work schedules, and corridor-level capacity protection. By contrast, the relatively higher temporal elasticity and route flexibility observed during the evening peak imply greater potential for adaptive system responses. In this context, resilience can be enhanced through operational traffic management strategies, including dynamic signal control, route guidance, and corridor-specific interventions that facilitate demand redistribution. This distinction indicates that effective resilience enhancement requires time-specific and mechanism-aware control strategies, rather than uniform peak-hour management approaches. Seasonal operating environments may further modulate these dynamics (e.g., winter conditions may reduce effective capacity and increase travel time sensitivity), thereby amplifying performance degradation under comparable demand conditions.

Overall, daily operational resilience is condition-dependent: weekday and weekend regimes exhibit systematically different demand organization and corridor-level stress concentrations, implying recurrent "vulnerability windows" that can be targeted via time-of-day control and corridor-level management.

### 5.2. Mechanisms underlying resilience evolution: performance-structure interplay

Beyond characterizing aggregate resilience levels, this study further reveals the underlying mechanisms of resilience evolution at the link level. The proposed vector-based representation decomposes resilience variation into two first-order components: changes in link operating performance and changes in demand-induced structural dependence. Compared with Hamedmoghadam et al. (2021), which focuses on demand-aware percolation for bottleneck identification and network-wide transitions, this study extends the framework to time-window resilience and link-level trend decomposition under daily disturbances. Based on this approach, four typical resilience evolution patterns are identified, including a representative phenomenon referred to as "high-pressure recovery".

Specifically, during morning and evening peak periods, although network-level resilience declines markedly, a subset of road links exhibits improving resilience trends, indicating the presence of a stage-wise adaptive adjustment process under high operational pressure. When critical corridors approach capacity limits, marginal efficiency on core links declines and demand begins to spill over. Parallel roads and alternative routes with residual capacity begin to accommodate overflow demand. Under high-pressure conditions, these links perform a redundancy function and exhibit temporary improvements in local operating performance. Meanwhile, travelers' route choice adjustments and navigation-assisted rerouting are consistent with a faster redistribution of demand toward non-core links. Local traffic management (e.g., signal timing adjustments) may further facilitate this shift. Similar phenomena involving traffic redistribution toward parallel or suboptimal routes under high-load conditions have been reported in the context of network efficiency and flow rebalancing (Youn et al., 2008; Gao et al., 2012). However, such studies primarily interpret these dynamics from efficiency or flow perspectives and have not systematically incorporated them into a resilience evolution framework. Accordingly, "high-pressure recovery" can be understood as a transitional stage in the resilience evolution process under daily traffic disturbances, reflecting the endogenous regulatory capacity of urban road transportation systems under extreme routine pressure. This phenomenon does not imply immediate system-wide recovery, but rather reflects localized adaptive responses that precede observable improvements at the global scale.

This distinction between resilience "states" and "evolutionary trends" provides a dual perspective that better captures the nonlinearity and path dependence of urban traffic systems, where recovery processes often emerge locally before becoming observable at the network scale.

### 5.3. Implications

The findings of this study yield several implications for urban traffic management aimed at improving operational efficiency while reducing the environmental burden associated with daily congestion. The proposed framework supports implications at two complementary levels. At the network level, the demand percolation curve $DWC_{\rho}^{t}$ and the temporally aggregated network-level resilience measure summarize overall accessibility retention and enable cross-period benchmarking (e.g., weekday/weekend and peak/off-peak contrasts). At the link level, the link weighted coefficient $lwc_{ij}^{t}$ and the evolution indicators (Section 5.2) reveal where demand-weighted structural dependence concentrates and how resilience contributions shift over time (Table 3). Because the two perspectives are mathematically consistent (Eq. (15)), the network-level benchmarking and link-level localization jointly motivate the following implications.

(1) The pronounced temporal variability of daily resilience suggests that uniform, static traffic management strategies are insufficient. Peak-period resilience degradation is primarily driven by concentrated and directional demand, indicating that time-specific interventions, such as adaptive signal control, corridor-based demand management, and staggered work schedules, are likely to be more effective than network-wide measures.
(2) The identification of localized recovery trends during periods of low network resilience highlights opportunities for proactive management. By reinforcing emerging alternative routes and alleviating secondary bottlenecks, traffic authorities can accelerate system-wide recovery and shorten the duration of inefficient operating states. Such selective interventions are particularly relevant for recurrently overburdened corridors that repeatedly exhibit high structural importance.
(3) Enhancing daily operational resilience has direct implications for cleaner production and environmental sustainability. Prolonged congestion under routine conditions contributes significantly to excessive fuel consumption, emissions, and energy waste. Improving resilience under everyday disturbances can therefore reduce cumulative environmental impacts more effectively than strategies focused solely on rare extreme events. From this perspective, daily resilience assessment provides a valuable analytical foundation for aligning traffic management practices with cleaner production objectives by reducing recurrent, system-wide congestion exposure. By identifying when and where congestion exposure concentrates, the framework supports actionable operational measures, such as idling reduction through signal coordination, corridor management, and demand management, to curb cumulative fuel use and emissions.

(4) The link-level evolution analysis offers a practical tool for prioritizing interventions. By distinguishing performance-driven and structure-driven resilience changes, planners can tailor strategies to specific mechanisms, such as improving traffic flow efficiency on performance-limited links or redistributing demand away from structurally overburdened corridors. Specifically, performance-driven deterioration motivates efficiency-oriented controls (signal coordination and rapid incident response), whereas structure-driven overburden calls for demand reallocation and access/route management on structurally critical corridors.

## 6. Conclusions

This study proposes a demand-aware, percolation-based framework to assess the daily operational resilience of URTS networks. By jointly incorporating link operating performance and demand-induced structural dependence within a unified analytical framework, the proposed approach enables resilience assessment from both network-level and link-level perspectives. In addition, a vector-based method for characterizing resilience evolution trends is proposed, allowing the direction and magnitude of resilience change to be explicitly quantified at the link level.

Using high-resolution traffic and demand data from Nanjing, the empirical analysis yields two key findings. First, daily operational resilience exhibits clear and systematic temporal variability across weekdays, pure weekends, peak periods and seasonal conditions, highlighting the limitations of static or single-time resilience indicators under recurrent traffic disturbances. Second, the link-level evolution analysis reveals that resilience dynamics are governed by the interplay between performance degradation and demand-driven structural reconfiguration, indicating that adaptive processes may emerge locally even when overall network resilience remains low. Because the required inputs are routinely available in many cities, the proposed assessment framework is transferable and can support comparative, cross-city benchmarking of daily operational resilience and related sustainability implications.

By extending percolation theory to daily urban road operations, the framework captures congestion-driven functional degradation and resilience evolution trends. In practice, this information can inform targeted interventions such as corridor-based demand management, signal timing optimization, and bottleneck-oriented capacity management to mitigate the network-wide impacts of recurrent congestion. Enhancing resilience under daily operating conditions also has important implications for sustainable urban mobility. Since recurrent congestion is a major contributor to cumulative energy consumption and emissions in urban transport systems, reducing the duration and spread of congestion under daily disturbances can improve operational efficiency and lower environmental burdens system-wide. From this perspective, daily resilience assessment offers a meaningful analytical foundation for aligning traffic management strategies with cleaner production objectives and broader sustainability goals.

However, there are still a few limitations in this research. First, the resilience measures rely on historical traffic operation data and are therefore intended for retrospective evaluation rather than real-time deployment. Second, the empirical analysis is based on a single metropolitan case, and the transferability of the findings to other cities with different network structures and travel behaviors requires further investigation. Third, emissions are discussed through operational implications but are not explicitly modeled within the resilience metric. Future research may extend this framework by incorporating rolling or adaptive baselines for near-real-time resilience monitoring, integrating traveler behavioral models and multi-modal interactions, and applying the approach to multiple cities to support comparative analysis of daily resilience patterns. Additionally, future work will integrate OD completion ratios and arrival-time reliability into the resilience assessment to better link snapshot-based network states with realized travel demand.

## CRediT authorship contribution statement

**Qi Ai:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Chengcheng Xu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Hao Tong:** Writing – review & editing, Data curation, Conceptualization. **Haosong Wen:** Writing – review & editing, Investigation, Formal analysis. **Chang Peng:** Software, Data curation, Conceptualization. **Michael Levin:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

## Code availability

Source codes for the algorithms proposed in this study are available at https://github.com/Nor-Eich/Daily-resilience-of-urban-road-transportation-systems-assessment-enhancement-and-recovery. Specific codes that produce the results presented in this paper are available upon request.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This research is supported by the National Natural Science Foundation of China (Grant No. 52525204, Grant No. 52172343, and Grant No. 52232012), Science and Technology Innovation Program of Xiongan New Area (Grant No.2023XAGG0089) and Hebei Provincial Science and Technology Research Program (No. 252F0802D).

## Appendix A. The advance of the proposed resilience assessment method

This research proposes a comprehensive resilience assessment framework that consists of both resilience assessment and resilience evolution tendency classification. It provides insights into the evolution of URTS resilience, progressing from the link level to the network level. The method for assessing resilience is a performance-based approach that aligns with the framework outlined in Liu et al. (2025). However, unlike those studies, the proposed method not only considers the performance of the system but also incorporates the characteristics of network topology, integrating both aspects to accurately capture changes in URTS.

Moreover, beyond the aforementioned reasons that enable the proposed method to accurately assess network resilience, the advantages of this approach are further discussed from a probabilistic perspective.

Let the random variable $X^t$ represent the set of qualitites for limiting links experienced during a random trip at time $t$, which consist of the instantaneous link performance $q_{ij}^t$. Thus, the probability that this random trip uses a specific limiting link $(i,j)$ is equal to the link weighted coefficient $lwc_{ij}^t$, specifically:

$$P\left\{X^t = q_{ij}^t\right\} = lwc_{ij}^t, \sum_{(i,j)\in E} lwc_{ij}^t = 1 \tag{34}$$

Hence, the expected travel performance at time $t$ is

$$\mathbb{E}[X^t] = \sum_{(i,j)\in E} q_{ij}^t \cdot P\left\{X^t = q_{ij}^t\right\} = \sum_{(i,j)\in E} lwc_{ij}^t \cdot q_{ij}^t \tag{35}$$

which is identical to Eq. (12) and Eq. (13) in Section 3.3.3 and reflects the average performance that can be experienced during a random trip at time $t$, essentially represent the average performance of the network at that time. Treating $\mathbb{E}[X^t]$ as a time series over the disturbance window $[t_1, t_2]$ and taking a weighted temporal average yields the following equation

$$R = \mathbb{E}[X] = \frac{1}{t_2 - t_1} \sum_{t=t_1}^{t_2} \mathbb{E}[X^t] = \sum_{t=t_1}^{t_2} \sum_{(i,j)\in E} lwc_{ij}^t \cdot q_{ij}^t \tag{36}$$

which precisely matches the formulation given in Section 3.3. This derivation shows that the proposed indicator $R$ is the composite of a spatial expectation and a temporal expectation, rigorously integrating demand distribution and link performance; it therefore provides an accurate, theoretically grounded measure of network resilience.

## Appendix B. Sensitivity analysis of resilience evolutionary tendency classification

The proposed resilience evolutionary tendency classification is based on the sign structure of first-order variations in link operational performance and structural dependence. As such, it is necessary to assess whether the resulting classification is robust to realistic measurement noise and whether adopting a zero threshold for sign determination leads to noise-dominated or unstable outcomes.

This sensitivity analysis is designed to address three specific concerns: (1) Whether the four-quadrant classification is sensitive to realistic measurement uncertainty. (2) Whether using a zero threshold is arbitrary or unstable. (3) How robustness and coverage trade off when uncertainty buffering is introduced. To this end, a joint noise-threshold analysis is conducted, in which measurement noise and classification thresholds are systematically varied and their combined effects are examined.

### *B.1 Noise modeling at the speed level*

Let $v_{ij}^t$ denote the observed average speed on link $(i,j)$ during time interval $t$. The perturbed speed is defined as:

$$\widetilde{v}_{ij}^t = v_{ij}^t \cdot (1+\epsilon), \epsilon \sim \mathcal{N}\left(0, \sigma^2\right),$$

where $\epsilon$ represents zero-mean Gaussian noise. The Gaussian assumption reflects the aggregation of multiple independent error sources in traffic sensing and temporal averaging and is widely adopted as a neutral and conservative approximation.

The magnitude of injected noise is controlled by a noise intensity parameter $\alpha$, which scales the dispersion of speed-induced indicators. Instead of using the standard deviation, the interquartile range (IQR) is employed to determine the noise scale (Lanzante, 1996):

$$\sigma_{\Delta q} = \alpha \cdot IQR(\Delta q), \sigma_{\Delta lwc} = \alpha \cdot IQR(\Delta lwc)$$

Noise levels $\alpha \in \{0.01, 0.02, 0.05, 0.10\}$ are examined, covering realistic detector uncertainty (1-2%) as well as higher stress-test scenarios.

### *B.2 Threshold-based uncertainty buffering*

To examine the sensitivity of sign-based classification near zero, a threshold buffer is introduced for both $\Delta q$ and $\Delta lwc$. For a given noise realization, thresholds are defined as:

$$\kappa_q = m \cdot \sigma_{\Delta q}, \kappa_{lwc} = m \cdot \sigma_{\Delta lwc},$$

where $m$ is a threshold multiplier. A link is labeled as UNCERTAIN if:

$$|\Delta q_{ij}^t| \le \kappa_q \text{ or } \left|\Delta lwc_{ij}^t\right| \le \kappa_{lwc}.$$

Otherwise, the link is classified into one of the four evolutionary tendency categories based on the signs of $\left(\Delta q_{ij}^t, \Delta lwc_{ij}^t\right)$. Threshold multipliers $m \in \{0.0, 0.25, 0.5, 1.0\}$ are considered. The case $m = 0$ corresponds to the baseline scenario with no uncertainty buffering, in which all links are forcibly classified.

### *B.3 Monte Carlo design and evaluation metrics*

For each combination of noise intensity $\alpha$ and threshold multiplier $m$, 200 Monte Carlo realizations are generated. Two complementary metrics are used to evaluate robustness:

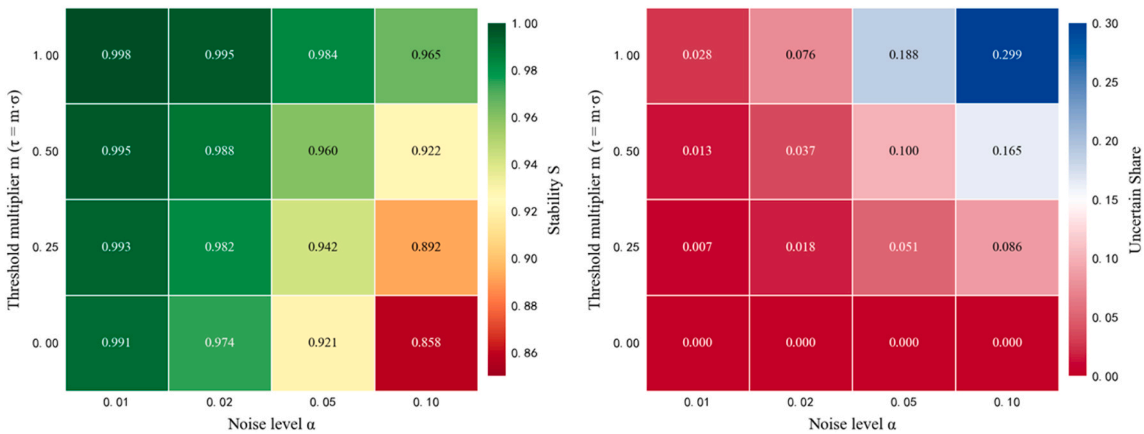
(1) Classification stability ($S$): Stability is defined as the proportion of links whose perturbed classification matches the baseline (noise-free) classification, conditional on both being non-UNCERTAIN:

$$S = \mathbb{P}\left(C_{ij}^{t,\mathrm{pert}} = C_{ij}^{t,\mathrm{base}} \,\middle|\, \text{both non-UNCERTAIN}\right).$$

(2) Uncertain share: The fraction of links classified as UNCERTAIN, representing the loss of coverage induced by threshold buffering.

Results are aggregated across all time periods to obtain mean stability and uncertainty ratios.

*B.4 Results: joint effects of noise and threshold*




(a) Stability under Noise × Threshold

(b) UNCERTAIN Share under Noise × Threshold

**Fig. 17.** Effects of noise and threshold (a) presents the heatmap of classification stability as a function of noise intensity $\alpha$ and threshold multiplier $m$. (b) presents the corresponding heatmap of the uncertain share.

The joint effects of noise and threshold are summarized in Fig. 17. Three consistent patterns emerge: (1) High robustness under realistic noise: For noise levels $\alpha \leq 0.02$, classification stability exceeds 97% even when no threshold buffering is applied ($m = 0$). This indicates that the four-quadrant evolutionary tendency classification is not dominated by realistic measurement noise. (2) Boundary-driven sensitivity under high noise: As noise increases ($\alpha = 0.05 - 0.10$), stability decreases when all samples are forcibly classified. This effect is concentrated among links with small absolute variations near zero, which are inherently sensitive to sign flips under perturbation. (3) Stability-coverage trade-off induced by buffering: Increasing $m$ systematically improves stability by excluding boundary samples from forced classification, at the cost of increasing the uncertain share. This confirms that improved stability arises from explicit uncertainty management rather than changes in the underlying classification logic.

This sensitivity analysis demonstrates that the observed classification patterns reflect mechanism-driven resilience evolution rather than noise artifacts. The zero threshold is therefore retained in the main analysis as a parsimonious reference choice, consistent with the first-order formulation of resilience variation. At the same time, the results clarify that alternative thresholds can be employed as a conservative option when data quality is poor or when analysts prefer to avoid classifying boundary cases. Importantly, introducing uncertainty buffering does not alter the qualitative structure of resilience evolutionary tendencies, but merely trades coverage for robustness.

## Data availability

22 days of Nanjing's road transportation network data used in this study, are available at https://doi.org/10.5281/zenodo.17218472.

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